

Data Sheet

**ENERCON SCADA
PDI-OPC**

Legal notice

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Document information

Document ID	D0141328-10		
Notation	Original document. Source document of this translation: D0138302-10.		
Date	Language	DCC	Plant / department
2014-10-21	eng	DA	WRD GmbH / Documentation Department

Table of contents

1	Introduction	5
2	Historical data	6
3	Nodes and items for data querying	8
3.1	The Loc node	8
3.2	The Loc/Wec node	9
3.2.1	The Loc/Wec/Plant<N> node	11
3.2.2	The Loc/Wec/Plant<N>/Status node	13
3.2.3	The Loc/Wec/Plant<N>/Log node	15
3.2.3.1	The Loc/Wec/Plant<N>/Log/Wec15 node (optional)	16
3.2.3.2	The Loc/Wec/Plant<N>/Log/Wec15/<Period> node (optional)	17
3.2.3.3	The Loc/Wec/Plant<N>/Log/T<XX> node	18
3.2.3.4	The Loc/Wec/Plant<N>/Log/T<XX>/<Period> node	19
3.2.3.5	The Loc/Wec/Plant<N>/Log/Wecstd node	21
3.2.3.6	The Loc/Wec/Plant<N>/Log/Wecstd/<Period> node	22
3.2.4	The Loc/Wec/Plant<N>/Ava node	24
3.2.4.1	The Loc/Wec/Plant<N>/Ava/<Period> node	26
3.2.5	The Loc/Wec/Plant<N>/Para node	27
3.3	The Loc/Sens node	48
3.3.1	The Loc/Sens/Plant<N> node	48
3.3.2	The Loc/Sens/Plant<N>/Log/Mspgerf1/<Period> node	49
3.4	The Loc/Conv node	50
3.4.1	The Loc/Conv/Plant<N> node	50
3.4.2	The Loc/Conv/Plant<N>/Log/Solar1/<Period> node	51
3.5	The Loc/Ctrl node	53
3.5.1	RTU	53
3.5.1.1	The Loc/Ctrl/Plant<N>/Log/Rtu1/<Period> node	55
3.5.1.2	The Loc/Ctrl/Plant<N>/Log/Rtu15/<Period> node (optional)	56
3.5.1.3	The Loc/Ctrl/Plant<N>/Log/Rtuphase1/<Period> node	57
3.5.1.4	The Loc/Ctrl/Plant<N>/Para for RTU node	58
3.5.2	GDA	65
3.5.2.1	The Loc/Ctrl/Plant<N>/Log/Farmctr1/<Period> node	67
3.5.2.2	The Loc/Ctrl/Plant<N>/Log/Farmctr15/<Period> node (optional)	69
3.5.3	FCU	70
3.5.3.1	The Loc/Ctrl/Plant<N>/Log/Fcuctr1/<Period> node	71
3.5.4	SCU	73
3.5.4.1	The Loc/Ctrl/Plant<N>/Log/Substat1/<Period> node	74

3.5.5	CLC	75
3.5.5.1	The Loc/Ctrl/Plant<N>/Log/Substat2/<Period> node	77
3.5.5.2	The Loc/Ctrl/Plant<N>/Log/SubTemp2/<Period> node	79
3.5.5.3	The Loc/Ctrl/Plant<N>/Log/Subphase2/<Period> node	80
3.6	The Loc/Meteo node	81
3.6.1	The Loc/Meteo/Plant<N> node	81
3.6.2	The Loc/Meteo/Plant<N>/Log/Wind1/<Period> node	82
3.6.3	The Loc/Meteo/Plant<Index>/Log/Weather2/<Period> node	83
3.7	The Loc/Actor node	84
3.7.1	PDI	84
3.7.1.1	The Loc/Actor/Plant<N>/Para node for PDI	85
4	Nodes and items for closed-loop and open-loop control	87
4.1	Session items for access for control purposes	87
4.1.1	Session items for access to individual WECs for control purposes	88
4.1.2	Session items for wind farm start/stop of all WECs in a wind farm	89
4.1.3	Session items for resetting the control system of individual wind energy converters	90
4.2	Items for wind farm closed-loop and open-loop control (ENERCON SCADA Server)	91
4.2.1	Weekly table control	91
4.2.2	Farm Control 2	93
4.2.3	Apparent power control 2	96
4.2.4	Open Loop Control (OICtr1)	98
4.2.5	Manual control 1 (ManCtr1)	101
4.2.6	Power Consumption Management	103
4.3	Items for RTU/FCU closed-loop and open-loop control	108
5	Overview of items and abbreviations used	113
5.1	Wind energy converters	113
5.2	ENERCON GDA	124
5.3	ENERCON METEO	126
5.4	ENERCON Type 1 SCU	128
5.5	ENERCON Type 2 SCU	128
5.6	ENERCON SCU with temperature measurement	131
5.7	ENERCON FCU	133
5.8	ENERCON RTU	135
5.9	Solar energy converter	137
5.10	Medium-voltage data acquisition	139
	Glossary of terms	140

1 Introduction

The ENERCON SCADA PDI-OPC is an OPC XML DA server in accordance with the V1.01 specifications of the OPC Foundation. Using the ENERCON SCADA PDI-OPC, wind farm data can be analysed online and control values can be transmitted to the wind farm without having to use the ENERCON SCADA Remote software.

This document describes the data tree und die OPC items of the ENERCON SCADA PDI-OPC.

This document describes both the standard scope of supply and the optional components. Which of the information and data included in this document can be called up and used by each customer depends on the systems installed in the farm.

2 Historical data

The following historical data is available via OPC:

Standard data (Fcuctr1, Farmctr1, Mspgerf1, Rtu1, Rtuphase1, Solar1, Substat1, Substat2, SubTemp2, Subphase2, Waecstd1 ,Wecstd, Wind1, Weather2)

Data type	Number of data sets	Recoding rate	Timeframe
Raw data (Raw)	720	60 s	12 hours
Report data (Rep)	288	600 s	48 hours
Daily data	61	1 day	60 days + average of current day
Weekly data	55	1 week	54 weeks + average of current week
Monthly data	37	1 month	36 months + average of current month
Yearly data	11	1 year	10 years + average of current year

15-minute data (Rtu15, Farmctr15) (optional)

Data type	Number of data sets	Recoding rate	Timeframe
Raw data (Raw)	720	60 s	12 hours
Report data (Rep)	288	900 s	72 hours

Availability (Ava)

Data type	Number of data sets	Recoding rate	Timeframe
Daily data	91	1 day	90 days + average of current day
Weekly data	25	1 week	24 weeks + average of current week
Monthly data	13	1 month	12 months + average of current month
Yearly data	11	1 year	10 years + average of current year

Temperature data (TXX)

Data type	Number of data sets	Recoding rate	Timeframe
Raw data (Raw)	60	60 s	1 hour
Report data (Rep.)	288	600 s	48 hours

Status data

PDI-OPC retains the 200 most current status messages.



If the ENERCON SCADA Server cannot retrieve any data from devices or equipment, e.g. due to a communication error, no data is available for this time period.

3 Nodes and items for data querying

In OPC, signs (+ and –) are used in accordance with IEC 61400-21.

3.1 The Loc node

This node contains the data of all WECs of a wind farm.

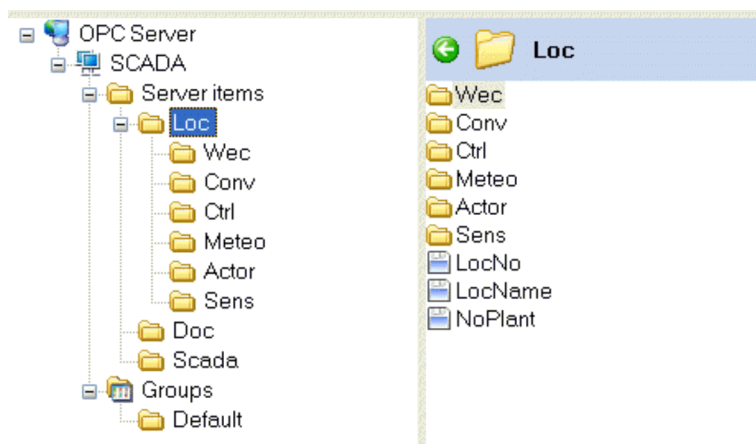


Fig. 1: *Loc* node

Node/Item	Description
Wec	Wind energy converter (WEC) data
Conv	Converter unit data (e.g. solar inverter)
Ctrl	Wind farm control system data (e.g. RTU, FCU)
Meteo	ENERCON METEO weather recording system data
Actor	Actor data (e.g. PDI)
Sens	Sensor unit data (e.g. medium-voltage acquisition)
LocNo	The wind farm number assigned by ENERCON
LocName	The wind farm name defined by the customer
NoPlant	Total number of all WECs in the wind farm

3.2 The Loc/Wec node

This node contains the data of the wind energy converters group.

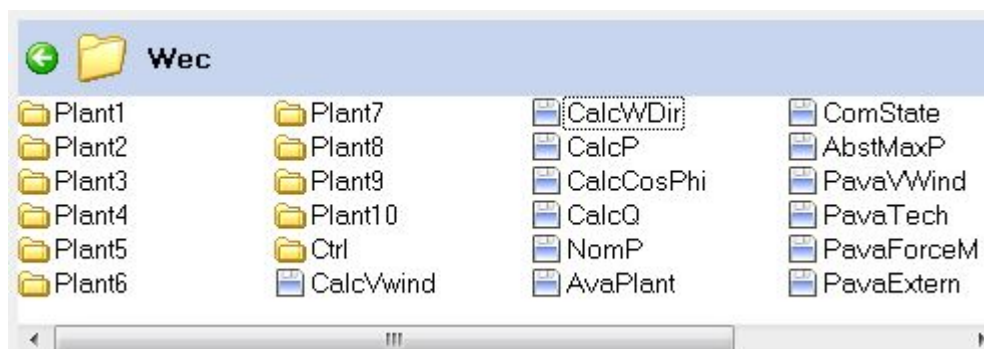


Fig. 2: Loc/Wec node

Node/Item	Description
Plant<N>	Data from wind energy converter <N>
Ctrl	Wind farm control (Start/Stop)
CalcVwind	Average wind speed across all wind energy converters [m/s]
CalcWDir	Average wind direction across all wind energy converters [°] (0° = North)
CalcP	Total power across all wind energy converters [kW]
CalcCosPhi	Average power factor $\cos \varphi$ across all wind energy converters
CalcQ	Total reactive power across all wind energy converters [kvar]
NomP	Nominal power [kW]
AvaPlant	Available WECs [%] (WECs that are able to inject power into the grid)
ComState	Available WEC communication [%] WEC communication will be considered inactive if three successive data query cycles have failed.
AbstMaxP	Abstract maximum active power
PavaVWind	Total theoretically available active power of wind energy converters in the wind [kW]. This value corresponds to the power the wind energy converters can feed into the grid in the current wind conditions, given the technical availability of all internal and external equipment and the absence of power output limitations through external setpoints.
PavaTech	Total technically available active power of wind energy converters in the wind [kW]. This value corresponds to the power the wind energy converters can provide in case of technical problems (generator overtemperature; defective converter, etc.). If there is no power limitation for technical reasons, this value equals the theoretically available active power PavaVWind.

Node/Item	Description
PavaForceM	Total of the maximum available active power of the wind energy converters, limited by force majeure [kW]. This value corresponds to the power the wind energy converters can provide in case of power limitations due to force majeure (storm control; grid overvoltage, etc.). If there is no power limitation due to force majeure, this value equals the theoretically available active power PavaVWind.
PavaExtern	Total of the available active power of the wind energy converters, limited by external setpoints [kW]. This value corresponds to the power the wind energy converters can provide in case of power limitations due to external setpoints. If there is no power limitation due to external setpoints, this value equals the theoretically available active power PavaVWind.

3.2.1 The Loc/Wec/Plant<N> node

This node contains all data of an individual wind energy converter.

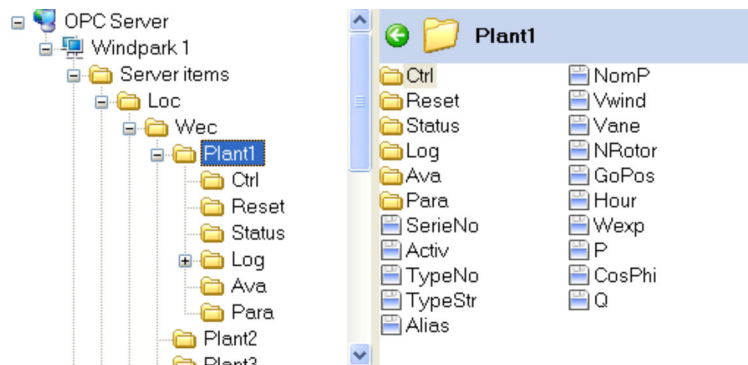


Fig. 3: Loc/Wec/Plant<N> node

Node/Item	Description
Ctrl	Control of this wind energy converter (Start/Stop)
Reset	Control system reset
Status	Status data of this wind energy converter
Log	Report data for this wind energy converter (minute, 10-minute, daily, weekly, etc.)
Ava	Availability data for this wind energy converter
Para	Parameters for this wind energy converter
SerieNo	ENERCON serial number
Active	WEC is active in the ENERCON SCADA System. In passive mode, the WEC is registered with the system but not considered with regard to any process. This mode is used for commissioning and during lengthy Service operations.
TypeNo	ENERCON control system type number
TypStr	ENERCON control system type string
Alias	Identification selectable by customer (10 digits)
NomP	Nominal power of wind energy converter [kW]
Vwind	Wind speed at wind energy converter [m/s]
Vane	Wind vane of wind energy converter (difference between wind direction and nacelle position) [°]
NRotor	Rotor speed of wind energy converter [rpm] (online value)
GoPos	Nacelle position of wind energy converter [°] (online value)
Hour	Operating hours counter of wind energy converter [h] (online value)
Wexp	Energy meter of wind energy converter [kWh] (online value)
P	Wind energy converter power [kW] (online value)
CosPhi	Power factor (cos ϕ) at wind energy converter (online value)
Q	Reactive power of wind energy converter [kvar] (online value)

Node/Item	Description
U-L1*	Low voltage U12 [V] (online value)
U-L2*	Low voltage U23 [V] (online value)
U-L3*	Low voltage U31 [V] (online value)
I-L1*	Current feed-in L1 [A] (online value)
I-L2*	Current feed-in L2 [A] (online value)
I-L3*	Current feed-in L3 [A] (online value)
Freq*	Grid frequency [Hz] (online value)
PavaVWind*	Theoretically available active power of the wind energy converter in the wind [kW]
PavaTech*	Technically available active power of the wind energy converter [kW]
PavaForceM	Maximum available active power of the wind energy converter, limited by force majeure [kW]
PavaExtern*	Maximum available active power of the wind energy converter, limited by external setpoints [kW]

*Optional

For a detailed description of the PavaVWind, PavaTech, PavaForceM and PavaExtern nodes, see The Loc/Wec node ch. 3.2, p. 9.



The number <N> of each WEC is assigned during commissioning and remains the same for that WEC's entire service life. The number cannot be changed because of its relation to data that are key to the ENERCON PartnerKonzept and warranties associated with it. Customers can use (and edit) the selected alias to individually identify the WEC.

3.2.2 The Loc/Wec/Plant<N>/Status node

The *Status* node contains current and historical statuses of the wind energy converter <N>. This data belongs to the report data group.

The node exists for all WEC ("Plant") groups. The following is an explanation of the node for one wind energy converter <N> only.

The current status as well as a history of past statuses can be displayed as an array of word. The maximum number of historical statuses stored in the ENERCON SCADA System is available in the *NoSt* item and is limited to a maximum of 200 statuses.

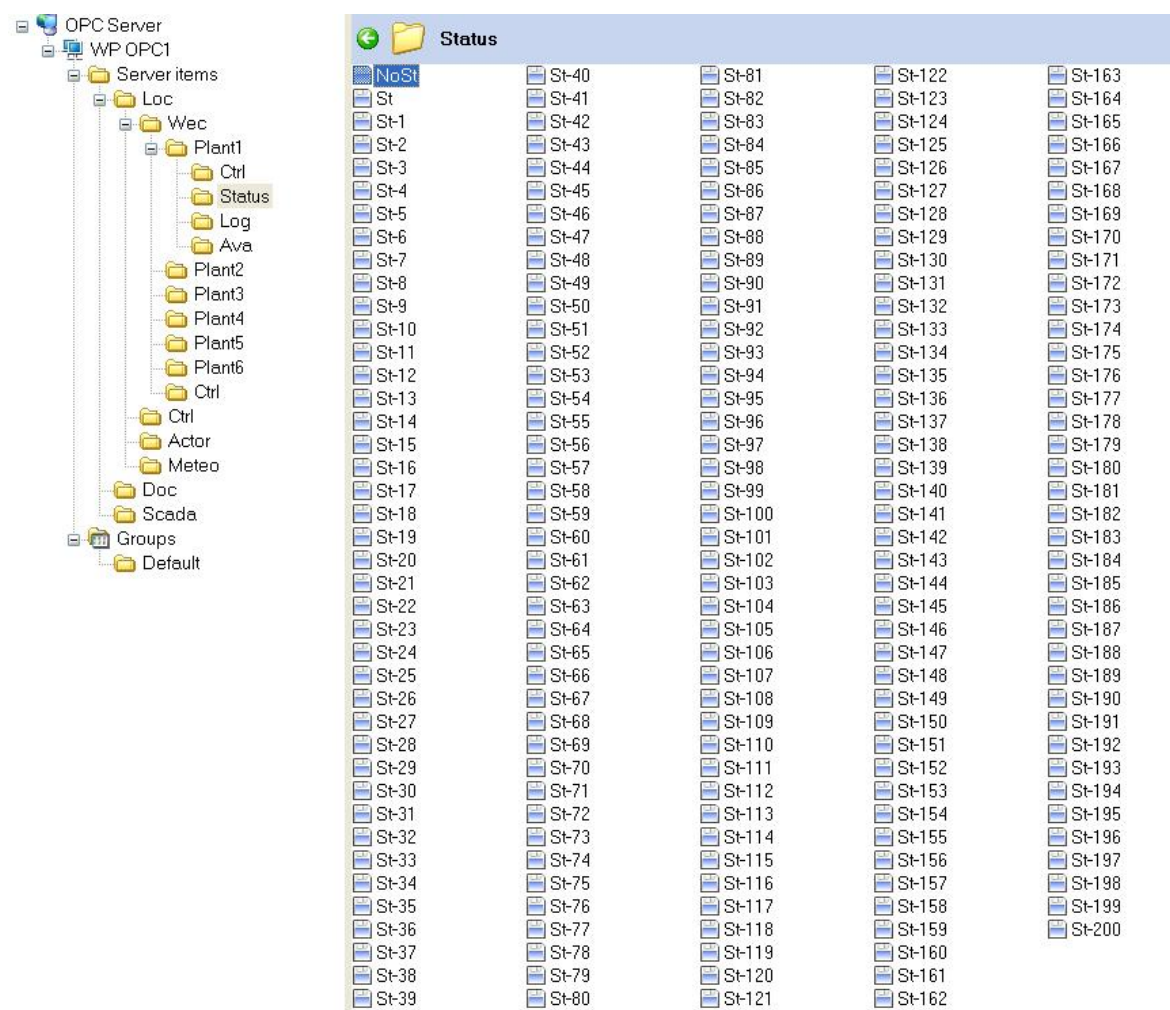


Fig. 4: The *Loc/Wec/Plant<N>/Status* node

Item	Description
NoSt	Number of available historical statuses (maximum: 200)
St	Current status
St-1	Historical status
St-2 to 199	Historical statuses (by date/time)

The elements of the array (ID 2) of the *Status* item have the meaning explained below:

Index	Description
0	Main status
1	Substatus
2	Fault message flag
3	Information
4	Additional information
5	Warning message flag
6	Service flag

A list of status messages for individual wind energy converter types is available from ENERCON Service IT on request.

3.2.3 The Loc/Wec/Plant<N>/Log node

The name of the *Log* node is derived from an abbreviation for Data Logger. This node contains report data for WEC <N> and their history. The node contains nodes for the WEC's standard data (*Wecstd*) as well as temperatures. If the temperature data is recorded by more than one data logger, the data loggers are shown in sequential order (e.g. *T82a1* and *T82a2* for systems with the CS82a control system).

The node contains all data that is also stored on the ENERCON SCADA Server as 10-minute, hourly, daily, etc. and as annual report data.

Optionally, the *Log* node can also contain selected 15-minute data (*Wec15* node).

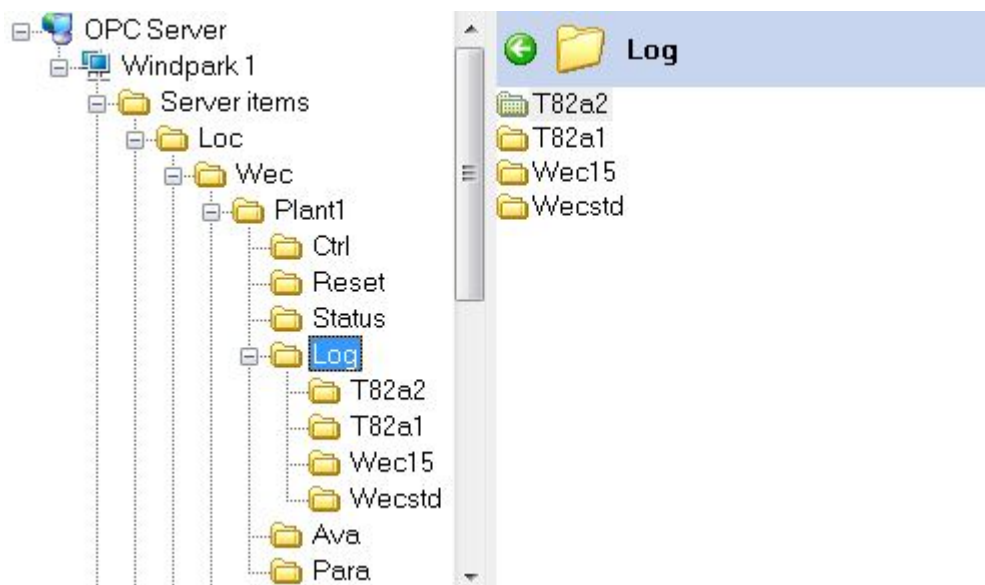


Fig. 5: *Loc/Wec/Plant<N>/Log* node

Node	Description
Wec15	15-minute data of the wind energy converter (optional)
T82	WEC temperatures T126a1, T126a2 and T126a3 = WECs with CS126a control system T101a1 and T101a2 = WECs with CS101a control system T82a1 and T82a2 = WECs with CS82a control system T70 = WECs with CS70a control system T70b = WECs with CS70b control system T66 = WECs with CS66a control system T48 = WECs with CS48a control system T44 = WECs with CS44a control system T44b = WECs with CS44b control system T40 = WECs with CS40a control system
Wecstd	Wind energy converter standard data

3.2.3.1 The Loc/Wec/Plant<N>/Log/Wec15 node (optional)

The optional *Loc/Wec/Plant<N>/Log/Wec15* node displays 15-minute data.

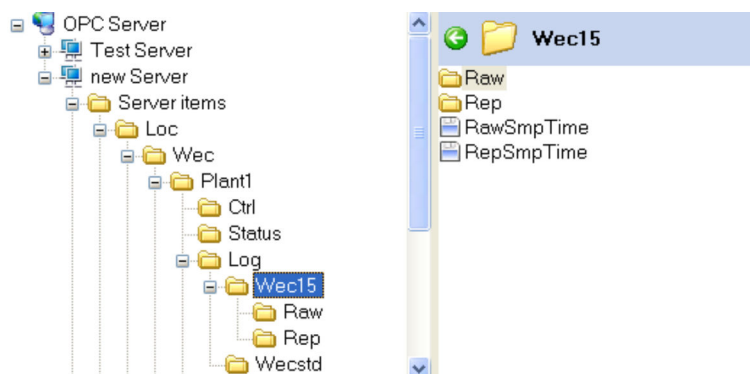


Fig. 6: The *Loc/Wec/Plant<N>/Log/Wec15* node

Node/Item	Description
Raw	Raw data (precalculated data)
Rep	Report data
RawSmpTime	Raw data sampling rate [s] (default: 60 s)
RepSmpTime	Report data sampling rate [s] (default: 900 s)

3.2.3.2 The Loc/Wec/Plant<N>/Log/Wec15/<Period> node (optional)

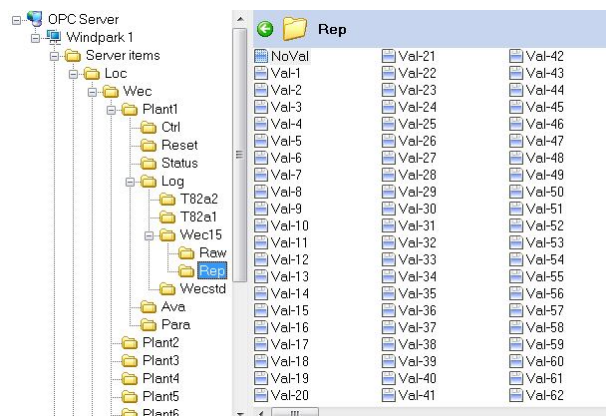


Fig. 7: The *Loc/Wec/Plant<N>/Log/Wec15/Raw* node (optional)

The *NoVal* item shows the number of data records. Depending on the node that contains the report data, the values of the *Val-<X>* item shown in an array (ID 2) have the meaning explained below.

Description	Interval	
	Raw	Rep
Number of sampling values	0	0
Average wind speed [m/s]	1	1
Average active power [kW]	2	2
Average nacelle position, including cable twisting [°]	3	3
Operating hours [h]	4	4
Energy production [kWh]	5	5
Average reactive power [kvar]	6	6

3.2.3.3 The Loc/Wec/Plant<N>/Log/T<XX> node

This *T<XX>* node contains temperature data for the wind energy converter. The name and the contents of the node depend on the control system type used (e.g. *T82a1* and *T82a2* for WECs with a CS82a control system).

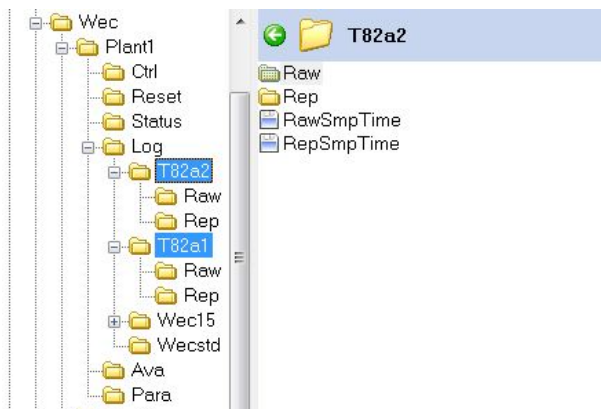


Fig. 8: The *Loc/Wec/Plant<N>/Log/T82a1* and *T82a2* nodes

Node/Item	Description
Raw	Raw data (precalculated data)
Rep	Report data
RawSmpTime	Raw data sampling rate [s] (default: 60 s)
RepSmpTime	Report data sampling rate [s] (default: 600 s)

3.2.3.4 The Loc/Wec/Plant<N>/Log/T<XX>/<Period> node

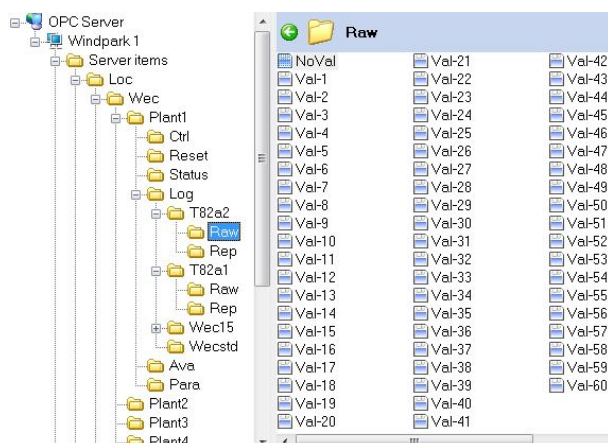


Fig. 9: The *Loc/Wec/Plant<N>/Log/T82a2/Raw* node

The *NoVal* item shows the number of data records. Depending on the node that contains the report data, the values of the *Val-<X>* item shown in an array (ID 2) have the meaning explained below.

CS82a (T82a1)

Description	Interval	
	Raw	Rep
Number of sampling values	0	0
Spinner temperature [°C]	1	1
Front bearing temperature [°C]	2	2
Rear bearing temperature [°C]	3	3
Blade A pitch heat sink [°C]	4	4
Blade B pitch heat sink [°C]	5	5
Blade C pitch heat sink [°C]	6	6
Blade A pitch control box [°C]	7	7
Blade B pitch control box [°C]	8	8
Blade C pitch control box [°C]	9	9
Blade A temperature [°C]	10	10
Blade B temperature [°C]	11	11
Blade C temperature [°C]	12	12
Rotor temperature 1 [°C]	13	13
Rotor temperature 2 [°C]	14	14
Stator temperature 1 [°C]	15	15
Stator temperature 2 [°C]	16	16
Nacelle ambient temperature [°C]	17	17
Nacelle temperature [°C]	18	18

Description	Interval	
	Raw	Rep
Nacelle control cabinet temperature [°C]	19	19
Rectifier 1 heat sink [°C]	20	20
Rectifier 2 heat sink [°C]	21	21
Rectifier cabinet temperature [°C]	22	22
Excitation heat sink [°C]	23	23
Ambient temperature [°C]	24	24
Tower temperature [°C]	25	25
Control cabinet temperature [°C]	26	26
Transformer temperature [°C]	27	27

CS82a (T82a2)

Description	Interval	
	Raw	Rep
Number of sampling values	0	0
Sys 1. Power cabinet 1 [°C]	1	1
Sys 1. Power cabinet 2 [°C]	2	2
Sys 1. Power cabinet 3 [°C]	3	3
Sys 1. Power cabinet 4 [°C]	4	4
Sys 1. Power cabinet 5 [°C]	5	5
Sys 1. Power cabinet 6 [°C]	6	6
Sys 1. Power cabinet 7 [°C]	7	7
Sys 1. Power cabinet 8 [°C]	8	8
Sys 1. Power cabinet 9 [°C]	9	9
Sys 1. Power cabinet 10 [°C]	10	10
Sys 1. Power cabinet 11 [°C]	11	11

3.2.3.5 The Loc/Wec/Plant<N>/Log/Wecstd node

The *Wecstd* node contains wind energy converter standard data.

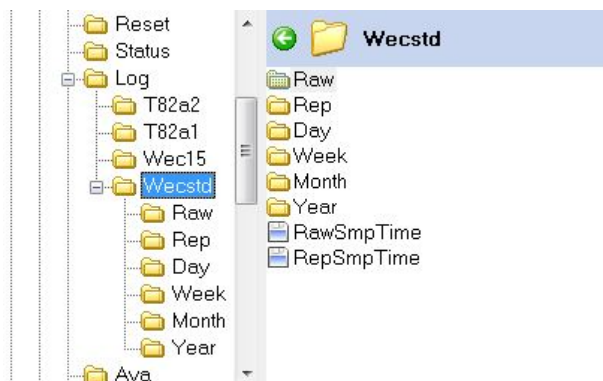


Fig. 10: The *Loc/Wec/Plant<N>/Log/Wecstd* node

Node/Item	Description
Raw	Raw data (precalculated WEC data)
Rep	Report data
Day	Daily reports
Week	Weekly reports
Month	Monthly reports
Year	Annual reports
RawSmpTime	Raw data sampling rate [s] (default: 60 s)
RepSmpTime	Report data sampling rate [s] (default: 600 s)

3.2.3.6 The Loc/Wec/Plant<N>/Log/Wecstd/<Period> node

The following is an explanation of the averages for one minute. The structure of 10-minute, daily, etc. and annual report data is identical.

The *Raw* node displays a list of the most recently computed averages for one minute and their history with regard to time. The first item (*NoVal*) shows the maximum number of historical values. This maximum number depends on the selected period (1 minute, 10 minutes, etc.) and can be looked up under *NoVal* for each value.

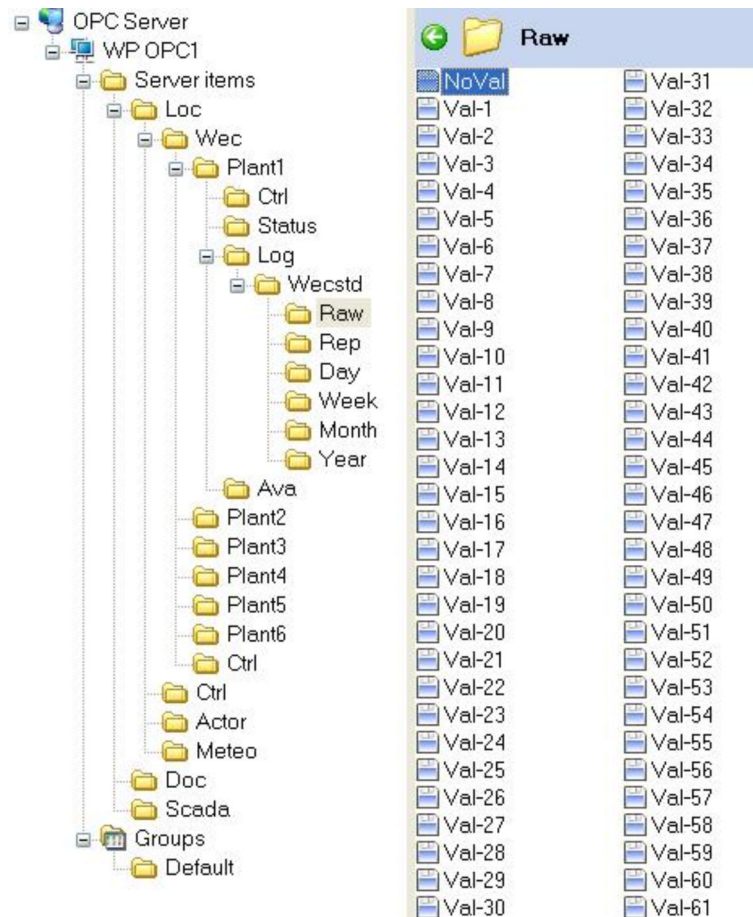


Fig. 11: The Loc/Wec/Plant<N>/Log/Wecstd/<Period> node

The *NoVal*/item shows the number of data records. Depending on the node that contains the report data, the values of the *Val-<X>* item shown in an array (ID 2) have the meaning explained below. If there is no numerical value, this measured value does not exist in that array.

Description of data record	Interval					
	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average wind speed [m/s]	1	1	1	1	1	1
Maximum wind speed [m/s]	2	2	2	2	2	2
Minimum wind speed [m/s]	3	3	3			
Average rotor speed [rpm]	4	4	4	3	3	3
Maximum rotor speed [rpm]	5	5	5	4	4	4
Minimum rotor speed [rpm]	6	6	6			
Average power [kW]	7	7	7	5	5	5
Maximum power [kW]	8	8	8	6	6	6
Minimum power [kW]	9	9	9			
Nacelle position [°]	10	10	10			
Operating hours [h]	11	11	11	7	7	7
Energy produced [kWh]	12	12	12	8	8	8
Productive minutes [min]	13	13	13	9	9	9
Average reactive power [kvar]	14	14	14	10	10	10
Maximum reactive power [kvar]	15	15	15	11	11	11
Minimum reactive power [kvar]	16	16	16	12	12	12
Average theoretically available active power in the wind [kW]*	17	17	17	13	13	13
Average technically available active power [kW]*	18	18	18	14	14	14
Average of the maximum available active power, limited by force majeure [kW]*	19	19	19	15	15	15
Average of the available active power, limited by external setpoints [kW]*	20	20	20	16	16	16
Average blade angle across A, B, C [°]	21	21	21			

* For a detailed description, see ch. 3.2, p. 9

3.2.4 The Loc/Wec/Plant<N>/Ava node

The *Ava* node contains report data about the availability of *Plant<N>* and its history.

The node exists for all WEC ("Plant") groups. The following is an explanation of the node for one wind energy converter <N> only.

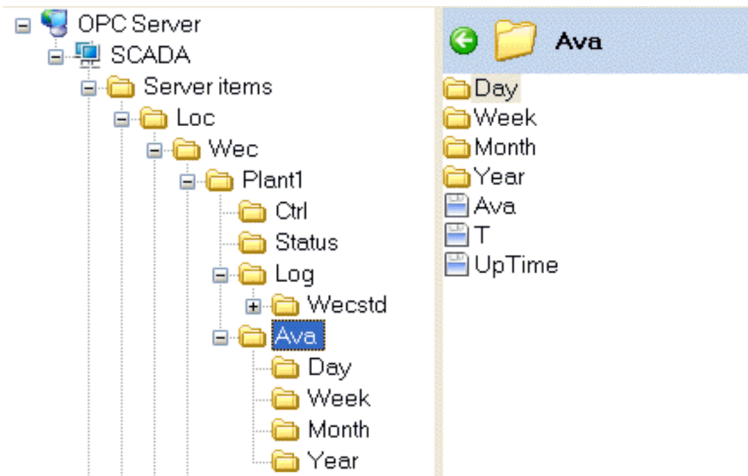


Fig. 12: The *Loc/Wec/Plant<N>/Ava* node

Node/Item	Description
Day	Daily availability
Week	Weekly availability
Month	Monthly availability
Year	Annual availability
Ava	Computed total availability [%] since commissioning of WEC <N>.
T	Array of T times as defined in the ENERCON PartnerKonzept (EPK).
UpTime	Start of computation of absolute availability. As a rule, the commissioning date of WEC <N>.

Index	Description
0	T1 [s]: Time during which the WEC is in operation or ready for operation, the WEC connections are available, no faults are displayed and the remote monitoring system is working.
1	T2 [s]: Total time since the start of availability measurement
2	T3 [s]: Time during which WEC connections are not available. A WEC connection is unavailable if connection parameters are out of the permissible range. Parameters in the utility grid are grid voltage and grid frequency at the WEC. Parameters are also out of the permissible range in the event of grid faults not caused by the WEC.

Index	Description
3	T4 [s]: Time during which the WEC is not in operation for the following reasons: ■ WEC has been put out of operation by customer or at customer's request due to modifications, parameterisation, measurements, maintenance work, or shutdown for operational reasons (except repairs). ■ Force majeure (Act of God) ■ Damage to WEC for which manufacturer is not responsible
4	T5 [s]: Time during which the remote monitoring system has failed or computed data have been lost
5	T6 [s]: Time during which the WEC displayed a fault and was therefore unavailable

Availability of WECs is computed as follows:

$$\frac{T1}{T2-T3-T4-T5}$$

3.2.4.1 The Loc/Wec/Plant<N>/Ava/<Period> node

The <Period> node contains availability data for historical time periods.

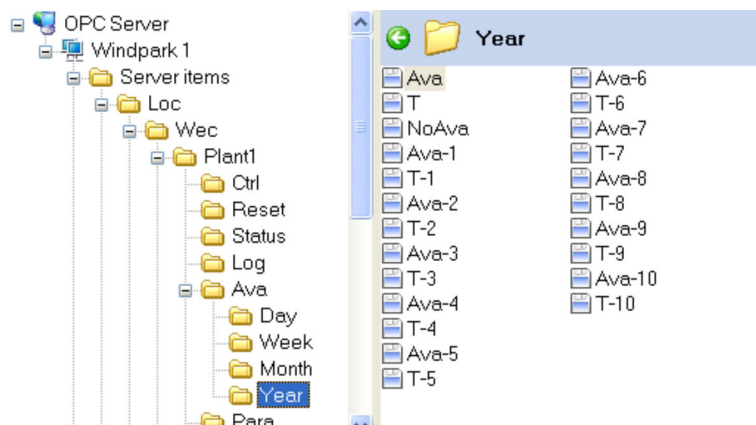


Fig. 13: The Loc/Wec/Plant<N>/Ava/<Period> node

Item	Description
Ava	Computed total availability [%] since commissioning of WEC <N>
T	Array of T times as defined in the ENERCON PartnerKonzept (EPK) (current time interval); see also ch. 3.2.4, p. 24
NoAva	Number of data records
Ava-1	Computed total availability [%] since commissioning of WEC <N> for the previous time interval
T-1	Array of T times as defined in the ENERCON PartnerKonzept (EPK) (current time interval); see also ch. 3.2.4, p. 24

3.2.5 The Loc/Wec/Plant<N>/Para node

The *Para* node can be used to view the control parameters of a wind energy converter or an ENERCON SCADA device. The item designation corresponds to the parameter designation in the control system or the ENERCON SCADA device.

The *Loc/Wec/Plant<N>/Para/User* node also includes items whose designations correspond to those of the parameters in the control system or the ENERCON SCADA device. The value of these items indicates who (user) made the last parameter change.

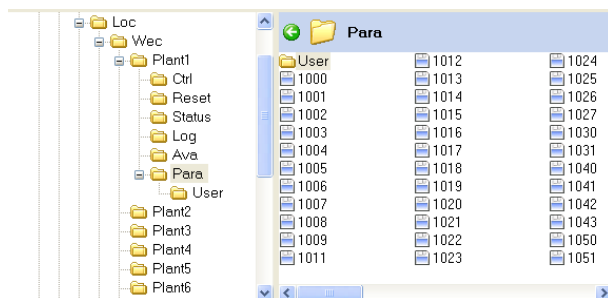


Fig. 14: The *Loc/Wec/Plant<N>/Para* node

CS48a

The following parameters are available for wind energy converters with a **CS48a** control system:

Item	Description	Unit
1000	Startdelay:	s
1001	nominal mains volt.:	V
1002	mains undervoltage level U<:	%
1003	tripping time during undervoltage U<:	s
1004	mains undervoltage level U<<:	%
1005	tripping time during undervoltage U<<:	s
1006	mains overvoltage level U>:	%
1007	tripping time during overvoltage U>:	s
1008	mains overvoltage level U>>:	%
1009	tripping time during overvoltage U>>:	s
1011	nominal mains freq.:	Hz
1012	mains underfrequency level:	Hz
1013	tripping time during underfrequency:	s
1014	mains overfrequency level:	Hz
1015	tripping time during overfrequency:	s
1016	frequency control:	
1017	frequency control referred to:	
1018	way of frequency control:	

Item	Description	Unit
1019	return frequency:	Hz
1020	limitation frequency 1:	Hz
1021	limitation frequency 2:	Hz
1022	limitation frequency 3:	Hz
1023	limitation frequency 4:	Hz
1024	limitation frequency 5:	Hz
1025	reserved power (f2-f3)	%
1026	limit. power (f4):	%
1027	limit. power (f5):	%
1030	grid type:	
1031	FACTS-operation:	
1040	mains undervoltage level for restart	%
1041	mains overvoltage level for restart	%
1042	mains underfrequency level for restart	Hz
1043	mains overfrequency level for restart	Hz
1050	inertia control:	
1051	inertia control:mode:	
1052	inertia trig. freq.:	Hz
1053	inertia min. freq.:	Hz
1054	inertia factor:	%
1055	max. inertia time:	s
1056	inertia recov. time:	s
1100	UVRT-mode during unsym. grid faults:	
1101	UVRT-mode during symmetr. grid faults:	
1102	UVRT trigger voltage:	%
1103	UVRT return voltage:	%
1104	UVRT trigger voltage ZPM:	%
1105	OVRT trigger voltage:	%
1106	OVRT return voltage:	%
1107	max. UVRT time during unsymmetr. faults:	s
1108	max. UVRT time during symmetr. faults:	s
1109	max. OVRT time:	s
1114	react. power gradient after ZPM:	MVA/s
1115	active power gradient after ZPM:	MW/s

Item	Description	Unit
1130	mains contactor during ZPM:	
1200	Statcom operation:	
1201	Farm Control Interf.:	
1202	apparent power limitation by:	
1203	v-curve gradient Q(rat) at P/P(rat):	%
1204	rated reactive power:	kvar
1205	reserved reactive power:	kvar
1209	dyn. react.pow.(Q/P): (farm contr. fault)	
1210	static react. pow.: (farm contr. faulty)	kvar
1308	SCADA farm control:	
1309	dP/dt-control:	
1311	storm control:	
1314	blade heating: automatic:	
1315	permit blade heating during operation:	
1316	max. power consump.of blade heating:	kW
1317	min.heating duration of blade heating:	min
1400	maximum power (farm contr. faulty):	kW
2000	site height above sealevel:	m
2003	nominal power:	
2010	cold climate package:	
2100	maximum power (power optimized):	kW
2101	maximum power (noise optimized):	kW
2102	power gradient (mains breakdown):	kW/s
2103	power gradient (during start):	kW/s
2104	power gradient (during operation):	kW/s
2501	nominal speed (noise optimized):	rpm
3101	Labko ice detector	
3102	Labko ice detector: automatic restart aft	
4011	sectoral limitation: start angle sector 1	°
4012	sectoral limitation: start angle sector 2	°
4013	sectoral limitation: start angle sector 3	°
4014	sectoral limitation: start angle sector 4	°
4015	sectoral limitation: start angle sector 5	°
4016	sectoral limitation: start angle sector 6	°

Item	Description	Unit
4017	sectoral limitation: start angle sector 7	°
4018	sectoral limitation: start angle sector 8	°
4021	sectoral limitation: end angle sector 1	°
4022	sectoral limitation: end angle sector 2	°
4023	sectoral limitation: end angle sector 3	°
4024	sectoral limitation: end angle sector 4	°
4025	sectoral limitation: end angle sector 5	°
4026	sectoral limitation: end angle sector 6	°
4027	sectoral limitation: end angle sector 7	°
4028	sectoral limitation: end angle sector 8	°
4031	sectoral limitation: min. wind speed for	m/s
4032	sectoral limitation: min. wind speed for	m/s
4033	sectoral limitation: min. wind speed for	m/s
4034	sectoral limitation: min. wind speed for	m/s
4035	sectoral limitation: min. wind speed for	m/s
4036	sectoral limitation: min. wind speed for	m/s
4037	sectoral limitation: min. wind speed for	m/s
4038	sectoral limitation: min. wind speed for	m/s
4041	sectoral limitation: max. wind speed for	m/s
4042	sectoral limitation: max. wind speed for	m/s
4043	sectoral limitation: max. wind speed for	m/s
4044	sectoral limitation: max. wind speed for	m/s
4045	sectoral limitation: max. wind speed for	m/s
4046	sectoral limitation: max. wind speed for	m/s
4047	sectoral limitation: max. wind speed for	m/s
4048	sectoral limitation: max. wind speed for	m/s
4051	sectoral limitation: max. power in sector	kW
4052	sectoral limitation: max. power in sector	kW
4053	sectoral limitation: max. power in sector	kW
4054	sectoral limitation: max. power in sector	kW
4055	sectoral limitation: max. power in sector	kW
4056	sectoral limitation: max. power in sector	kW
4057	sectoral limitation: max. power in sector	kW
4058	sectoral limitation: max. power in sector	kW

Item	Description	Unit
4061	sectoral limitation: min. blade angle in	°
4062	sectoral limitation: min. blade angle in	°
4063	sectoral limitation: min. blade angle in	°
4064	sectoral limitation: min. blade angle in	°
4065	sectoral limitation: min. blade angle in	°
4066	sectoral limitation: min. blade angle in	°
4067	sectoral limitation: min. blade angle in	°
4068	sectoral limitation: min. blade angle in	°
4091	sectoral limit.: power grad. sector 1-8	kW/s
4106	limit for farm ice detection:	%
4107	automatic restart after icing:	
4108	automatic restart during icing:	
4109	position nacelle during icing up to 7m/s	
4110	nacelle position at icing:	°
6000	noise optimization	
6001	minimum time	min
6002	noise optimization time	
6003	time from	hh:mm
6004	time to	hh:mm
6005	noise optimization wind speed	
6006	Delay	min
6007	min. wind speed	m/s
6008	max. wind speed	m/s
6009	noise optimization wind direction	
6010	delay	min
6011	angle 1. sector	°
6012	range 1. sector	°
6013	angle 2. sector	°
6014	range 2. sector	°
6015	angle 3. sector	°
6016	range 3. sector	°
6017	combination	
6100	shadow shut off	
6101	Shut off intensity	%

Item	Description	Unit
6102	filter on	s
6103	filter off	s

CS82a

The following parameters are available for wind energy converters with a **CS82a** control system:

Item	Description	Unit
1000	Startdelay:	s
1001	nominal mains volt.:	V
1002	mains undervoltage level U<:	%
1003	tripping time during undervoltage U<:	s
1004	mains undervoltage level U<<:	%
1005	tripping time during undervoltage U<<	s
1006	mains overvoltage level U>:	%
1007	tripping time during overvoltage U>:	s
1008	mains overvoltage level U>>:	%
1009	tripping time during overvoltage U>>:	s
1011	nominal mains freq.:	Hz
1012	mains underfrequency level:	Hz
1013	tripping time during underfrequency:	s
1014	mains overfrequency level:	Hz
1015	tripping time during overfrequency:	s
1016	frequency control:	
1017	frequency control referred to:	
1018	way of frequency control:	
1019	return frequency:	Hz
1020	limitation frequency 1:	Hz
1021	limitation frequency 2:	Hz
1022	limitation frequency 3:	Hz
1023	limitation frequency 4:	Hz
1024	limitation frequency 5:	Hz
1025	reserved power (f2-f3)	%
1026	limit. power (f4):	%
1027	limit. power (f5):	%
1030	grid type:	
1031	FACTS-operation:	
1040	mains undervoltage level for restart	%
1041	mains overvoltage level for restart	%
1042	mains underfrequency level for restart	Hz

Item	Description	Unit
1043	mains overfrequency level for restart	Hz
1050	inertia control:	
1051	inertia control:mode:	
1052	inertia trig. freq.:	Hz
1053	inertia min. freq.:	Hz
1054	inertia factor:	%
1055	max. inertia time:	s
1056	inertia recov. time:	s
1100	UVRT-mode during unsym. grid faults:	
1101	UVRT-mode during symmetr. grid faults:	
1102	UVRT trigger voltage:	%
1103	UVRT return voltage:	%
1104	UVRT trigger voltage ZPM:	%
1105	OVRT trigger voltage:	%
1106	OVRT return voltage:	%
1107	max. UVRT time during unsymmetr. faults:	s
1108	max. UVRT time during symmetr. faults:	s
1109	max. OVRT time:	s
1114	react. power gradient after ZPM:	MVar/s
1115	active power gradient after ZPM:	MW/s
1130	mains contactor during ZPM:	
1200	Statcom operation:	
1201	Farm Control Interf.:	
1202	apparent power limitation by:	
1203	v-curve gradient Q(rat) at P/P(rat):	%
1204	rated reactive power:	kvar
1205	reserved reactive power:	kvar
1209	dyn. react.pow.(Q/P): (farm contr. fault)	
1210	static react. pow.: (farm contr. faulty)	kvar
1308	SCADA farm control:	
1309	dP/dt-control:	
1311	storm control:	
1314	blade heating: automatic:	
1315	permit blade heating during operation:	

Item	Description	Unit
1316	max. power consump.of blade heating:	kW
1317	min. heating duration of blade heating:	Min
1400	maximum power (farm contr. faulty):	kW
2000	site height above sealevel:	m
2003	nominal power:	
2010	cold climate package:	
2100	maximum power (power optimized):	kW
2101	maximum power (noise optimized):	kW
2102	power gradient (mains breakdown):	kW/s
2103	power gradient (during start):	kW/s
2104	power gradient (during operation):	kW/s
2501	nominal speed (noise optimized):	rpm
3101	Labko ice detector	
3102	Labko ice detector: automatic restart aft	
4011	sectoral limitation: start angle sector 1	°
4012	sectoral limitation: start angle sector 2	°
4013	sectoral limitation: start angle sector 3	°
4014	sectoral limitation: start angle sector 4	°
4015	sectoral limitation: start angle sector 5	°
4016	sectoral limitation: start angle sector 6	°
4017	sectoral limitation: start angle sector 7	°
4018	sectoral limitation: start angle sector 8	°
4021	sectoral limitation: end angle sector 1	°
4022	sectoral limitation: end angle sector 2	°
4023	sectoral limitation: end angle sector 3	°
4024	sectoral limitation: end angle sector 4	°
4025	sectoral limitation: end angle sector 5	°
4026	sectoral limitation: end angle sector 6	°
4027	sectoral limitation: end angle sector 7	°
4028	sectoral limitation: end angle sector 8	°
4031	sectoral limitation: min. wind speed for	m/s
4032	sectoral limitation: min. wind speed for	m/s
4033	sectoral limitation: min. wind speed for	m/s
4034	sectoral limitation: min. wind speed for	m/s

Item	Description	Unit
4035	sectoral limitation: min. wind speed for	m/s
4036	sectoral limitation: min. wind speed for	m/s
4037	sectoral limitation: min. wind speed for	m/s
4038	sectoral limitation: min. wind speed for	m/s
4041	sectoral limitation: max. wind speed for	m/s
4042	sectoral limitation: max. wind speed for	m/s
4043	sectoral limitation: max. wind speed for	m/s
4044	sectoral limitation: max. wind speed for	m/s
4045	sectoral limitation: max. wind speed for	m/s
4046	sectoral limitation: max. wind speed for	m/s
4047	sectoral limitation: max. wind speed for	m/s
4048	sectoral limitation: max. wind speed for	m/s
4051	sectoral limitation: max. power in sector	kW
4052	sectoral limitation: max. power in sector	kW
4053	sectoral limitation: max. power in sector	kW
4054	sectoral limitation: max. power in sector	kW
4055	sectoral limitation: max. power in sector	kW
4056	sectoral limitation: max. power in sector	kW
4057	sectoral limitation: max. power in sector	kW
4058	sectoral limitation: max. power in sector	kW
4061	sectoral limitation: min. blade angle in	°
4062	sectoral limitation: min. blade angle in	°
4063	sectoral limitation: min. blade angle in	°
4064	sectoral limitation: min. blade angle in	°
4065	sectoral limitation: min. blade angle in	°
4066	sectoral limitation: min. blade angle in	°
4067	sectoral limitation: min. blade angle in	°
4068	sectoral limitation: min. blade angle in	°
4091	sectoral limit.: power grad. sector 1-8	kW/s
4106	limit for farm ice detection:	%
4107	automatic restart after icing:	
4108	automatic restart during icing:	
4109	position nacelle during icing up to 7m/s	
4110	nacelle position at icing:	°

Item	Description	Unit
6000	noise optimization	
6001	minimum time	min
6002	noise optimization time	
6003	time from	hh:mm
6004	time to	hh:mm
6005	noise optimization wind speed	
6006	delay	min
6007	min. wind speed	m/s
6008	max. wind speed	m/s
6009	noise optimization wind direction	
6010	delay	min
6011	angle 1. sector	°
6012	range 1. sector	°
6013	angle 2. sector	°
6014	range 2. sector	°
6015	angle 3. sector	°
6016	range 3. sector	°
6017	combination	
6100	shadow shut off	
6101	Shut off intensity	%
6102	filter on	s
6103	filter off	s

CS101a

The following parameters are available for wind energy converters with a **CS101a** control system:

Item	Description	Unit
1000	Startdelay:	s
1001	nominal mains volt.:	V
1002	mains undervoltage level U<:	%
1003	tripping time during undervoltage U<:	s
1004	mains undervoltage level U<<:	%
1005	tripping time during undervoltage U<<	s
1006	mains overvoltage level U>:	%
1007	tripping time during overvoltage U>:	s
1008	mains overvoltage level U>>:	%
1009	tripping time during overvoltage U>>:	s
1011	nominal mains freq.:	Hz
1012	mains underfrequency level:	Hz
1013	tripping time during underfrequency:	s
1014	mains overfrequency level:	Hz
1015	tripping time during overfrequency:	s
1016	frequency control:	
1017	frequency control referred to:	
1018	way of frequency control:	
1019	return frequency:	Hz
1020	limitation frequency 1:	Hz
1021	limitation frequency 2:	Hz
1022	limitation frequency 3:	Hz
1023	limitation frequency 4:	Hz
1024	limitation frequency 5:	Hz
1025	reserved power (f2-f3)	%
1026	limit. power (f4):	%
1027	limit. power (f5):	%
1030	grid type:	
1031	FACTS-operation:	
1040	mains undervoltage level for restart	%
1041	mains overvoltage level for restart	%
1042	mains underfrequency level for restart	Hz

Item	Description	Unit
1043	mains overfrequency level for restart	Hz
1050	inertia control:	
1051	inertia control:mode:	
1052	inertia trig. freq.:	Hz
1053	inertia min. freq.:	Hz
1054	inertia factor:	%
1055	max. inertia time:	s
1056	inertia recov. time:	s
1100	UVRT-mode during unsym. grid faults:	
1101	UVRT-mode during symmetr. grid faults:	
1102	UVRT trigger voltage:	%
1103	UVRT return voltage:	%
1104	UVRT trigger voltage ZPM:	%
1105	OVRT trigger voltage:	%
1106	OVRT return voltage:	%
1107	max. UVRT time during unsymmetr. faults:	s
1108	max. UVRT time during symmetr. faults:	s
1109	max. OVRT time:	s
1114	react. power gradient after ZPM:	MVar/s
1115	active power gradient after ZPM:	MW/s
1130	mains contactor during ZPM:	
1200	Statcom operation:	
1201	Farm Control Interf.:	
1202	apparent power limitation by:	
1203	v-curve gradient Q(rat) at P/P(rat):	%
1204	rated reactive power:	kvar
1205	reserved reactive power:	kvar
1209	dyn. react.pow.(Q/P): (farm contr. fault)	
1210	static react. pow.: (farm contr. faulty)	kvar
1308	SCADA farm control:	
1309	dP/dt-control:	
1311	storm control:	
1314	blade heating: automatic:	
1315	permit blade heating during operation:	

Item	Description	Unit
1316	max. power consump.of blade heating:	kW
1317	min.heating duration of blade heating:	min
1400	maximum power (farm contr. faulty):	kW
2000	site height above sealevel:	m
2003	nominal power:	
2010	cold climate package:	
2100	maximum power (power optimized):	kW
2101	maximum power (noise optimized):	kW
2102	power gradient (mains breakdown):	kW/s
2103	power gradient (during start):	kW/s
2104	power gradient (during operation):	kW/s
2501	nominal speed (noise optimized):	rpm
3101	Labko ice detector	
3102	Labko ice detector: automatic restart aft	
4011	sectoral limitation: start angle sector 1	°
4012	sectoral limitation: start angle sector 2	°
4013	sectoral limitation: start angle sector 3	°
4014	sectoral limitation: start angle sector 4	°
4015	sectoral limitation: start angle sector 5	°
4016	sectoral limitation: start angle sector 6	°
4017	sectoral limitation: start angle sector 7	°
4018	sectoral limitation: start angle sector 8	°
4021	sectoral limitation: end angle sector 1	°
4022	sectoral limitation: end angle sector 2	°
4023	sectoral limitation: end angle sector 3	°
4024	sectoral limitation: end angle sector 4	°
4025	sectoral limitation: end angle sector 5	°
4026	sectoral limitation: end angle sector 6	°
4027	sectoral limitation: end angle sector 7	°
4028	sectoral limitation: end angle sector 8	°
4031	sectoral limitation: min. wind speed for	m/s
4032	sectoral limitation: min. wind speed for	m/s
4033	sectoral limitation: min. wind speed for	m/s
4034	sectoral limitation: min. wind speed for	m/s

Item	Description	Unit
4035	sectoral limitation: min. wind speed for	m/s
4036	sectoral limitation: min. wind speed for	m/s
4037	sectoral limitation: min. wind speed for	m/s
4038	sectoral limitation: min. wind speed for	m/s
4041	sectoral limitation: max. wind speed for	m/s
4042	sectoral limitation: max. wind speed for	m/s
4043	sectoral limitation: max. wind speed for	m/s
4044	sectoral limitation: max. wind speed for	m/s
4045	sectoral limitation: max. wind speed for	m/s
4046	sectoral limitation: max. wind speed for	m/s
4047	sectoral limitation: max. wind speed for	m/s
4048	sectoral limitation: max. wind speed for	m/s
4051	sectoral limitation: max. power in sector	kW
4052	sectoral limitation: max. power in sector	kW
4053	sectoral limitation: max. power in sector	kW
4054	sectoral limitation: max. power in sector	kW
4055	sectoral limitation: max. power in sector	kW
4056	sectoral limitation: max. power in sector	kW
4057	sectoral limitation: max. power in sector	kW
4058	sectoral limitation: max. power in sector	kW
4061	sectoral limitation: min. blade angle in	°
4062	sectoral limitation: min. blade angle in	°
4063	sectoral limitation: min. blade angle in	°
4064	sectoral limitation: min. blade angle in	°
4065	sectoral limitation: min. blade angle in	°
4066	sectoral limitation: min. blade angle in	°
4067	sectoral limitation: min. blade angle in	°
4068	sectoral limitation: min. blade angle in	°
4091	sectoral limit.: power grad. sector 1-8	kW/s
4106	limit for farm ice detection:	%
4107	automatic restart after icing:	
4108	automatic restart during icing:	
4109	position nacelle during icing up to 7m/s	
4110	nacelle position at icing:	°

Item	Description	Unit
6000	noise optimization	
6001	minimum time	min
6002	noise optimization time	
6003	time from	hh:mm
6004	time to	hh:mm
6005	noise optimization wind speed	
6006	delay	min
6007	min. wind speed	m/s
6008	max. wind speed	m/s
6009	noise optimization wind direction	
6010	delay	min
6011	angle 1. sector	°
6012	range 1. sector	°
6013	angle 2. sector	°
6014	range 2. sector	°
6015	angle 3. sector	°
6016	range 3. sector	°
6017	combination	
6100	shadow shut off	
6101	Shut off intensity	%
6102	filter on	s
6103	filter off	s

CS126a

The following parameters are available for wind energy converters with a **CS126a** control system:

Item	Description	Unit
1000	Startdelay:	s
1001	nominal mains volt.:	V
1002	mains undervoltage level U<:	%
1003	tripping time during undervoltage U<:	s
1004	mains undervoltage level U<<:	%
1005	tripping time during undervoltage U<<	s
1006	mains overvoltage level U>:	%
1007	tripping time during overvoltage U>:	s
1008	mains overvoltage level U>>:	%
1009	tripping time during overvoltage U>>:	s
1011	nominal mains freq.:	Hz
1012	mains underfrequency level:	Hz
1013	tripping time during underfrequency:	s
1014	mains overfrequency level:	Hz
1015	tripping time during overfrequency:	s
1016	frequency control:	
1017	frequency control referred to:	
1018	way of frequency control:	
1019	return frequency:	Hz
1020	limitation frequency 1:	Hz
1021	limitation frequency 2:	Hz
1022	limitation frequency 3:	Hz
1023	limitation frequency 4:	Hz
1024	limitation frequency 5:	Hz
1025	reserved power (f2-f3)	%
1026	limit. power (f4):	%
1027	limit. power (f5):	%
1030	grid type:	
1031	FACTS-operation:	
1040	mains undervoltage level for restart	%
1041	mains overvoltage level for restart	%
1042	mains underfrequency level for restart	Hz

Item	Description	Unit
1043	mains overfrequency level for restart	Hz
1050	inertia control:	
1051	inertia control:mode:	
1052	inertia trig. freq.:	Hz
1053	inertia min. freq.:	Hz
1054	inertia factor:	%
1055	max. inertia time:	s
1056	inertia recov. time:	s
1100	UVRT-mode during unsym. grid faults:	
1101	UVRT-mode during symmetr. grid faults:	
1102	UVRT trigger voltage:	%
1103	UVRT return voltage:	%
1104	UVRT trigger voltage ZPM:	%
1105	OVRT trigger voltage:	%
1106	OVRT return voltage:	%
1107	max. UVRT time during unsymmetr. faults:	s
1108	max. UVRT time during symmetr. faults:	s
1109	max. OVRT time:	s
1114	react. power gradient after ZPM:	MVar/s
1115	active power gradient after ZPM:	MW/s
1130	mains contactor during ZPM:	
1200	Statcom operation:	
1201	Farm Control Interf.:	
1202	apparent power limitation by:	
1203	v-curve gradient Q(rat) at P/P(rat):	%
1204	rated reactive power:	kvar
1205	reserved reactive power:	kvar
1209	dyn. react.pow.(Q/P): (farm contr. fault)	
1210	static react. pow.: (farm contr. faulty)	kvar
1308	SCADA farm control:	
1309	dP/dt-control:	
1311	storm control:	
1314	blade heating: automatic:	
1315	permit blade heating during operation:	

Item	Description	Unit
1316	max. power consump.of blade heating:	kW
1317	min.heating duration of blade heating:	min
1400	maximum power (farm contr. faulty):	kW
2000	site height above sealevel:	m
2003	nominal power:	
2010	cold climate package:	
2100	maximum power (power optimized):	kW
2101	maximum power (noise optimized):	kW
2102	power gradient (mains breakdown):	kW/s
2103	power gradient (during start):	kW/s
2104	power gradient (during operation):	kW/s
2501	nominal speed (noise optimized):	rpm
3101	Labko ice detector	
3102	Labko ice detector: automatic restart aft	
4011	sectoral limitation: start angle sector 1	°
4012	sectoral limitation: start angle sector 2	°
4013	sectoral limitation: start angle sector 3	°
4014	sectoral limitation: start angle sector 4	°
4015	sectoral limitation: start angle sector 5	°
4016	sectoral limitation: start angle sector 6	°
4017	sectoral limitation: start angle sector 7	°
4018	sectoral limitation: start angle sector 8	°
4021	sectoral limitation: end angle sector 1	°
4022	sectoral limitation: end angle sector 2	°
4023	sectoral limitation: end angle sector 3	°
4024	sectoral limitation: end angle sector 4	°
4025	sectoral limitation: end angle sector 5	°
4026	sectoral limitation: end angle sector 6	°
4027	sectoral limitation: end angle sector 7	°
4028	sectoral limitation: end angle sector 8	°
4031	sectoral limitation: min. wind speed for	m/s
4032	sectoral limitation: min. wind speed for	m/s
4033	sectoral limitation: min. wind speed for	m/s
4034	sectoral limitation: min. wind speed for	m/s

Item	Description	Unit
4035	sectoral limitation: min. wind speed for	m/s
4036	sectoral limitation: min. wind speed for	m/s
4037	sectoral limitation: min. wind speed for	m/s
4038	sectoral limitation: min. wind speed for	m/s
4041	sectoral limitation: max. wind speed for	m/s
4042	sectoral limitation: max. wind speed for	m/s
4043	sectoral limitation: max. wind speed for	m/s
4044	sectoral limitation: max. wind speed for	m/s
4045	sectoral limitation: max. wind speed for	m/s
4046	sectoral limitation: max. wind speed for	m/s
4047	sectoral limitation: max. wind speed for	m/s
4048	sectoral limitation: max. wind speed for	m/s
4051	sectoral limitation: max. power in sector	kW
4052	sectoral limitation: max. power in sector	kW
4053	sectoral limitation: max. power in sector	kW
4054	sectoral limitation: max. power in sector	kW
4055	sectoral limitation: max. power in sector	kW
4056	sectoral limitation: max. power in sector	kW
4057	sectoral limitation: max. power in sector	kW
4058	sectoral limitation: max. power in sector	kW
4061	sectoral limitation: min. blade angle in	°
4062	sectoral limitation: min. blade angle in	°
4063	sectoral limitation: min. blade angle in	°
4064	sectoral limitation: min. blade angle in	°
4065	sectoral limitation: min. blade angle in	°
4066	sectoral limitation: min. blade angle in	°
4067	sectoral limitation: min. blade angle in	°
4068	sectoral limitation: min. blade angle in	°
4091	sectoral limit.: power grad. sector 1-8	kW/s
4106	limit for farm ice detection:	%
4107	automatic restart after icing:	
4108	automatic restart during icing:	
4109	position nacelle during icing up to 7m/s	
4110	nacelle position at icing:	°

Item	Description	Unit
6000	noise optimization	
6001	minimum time	min
6002	noise optimization time	
6003	time from	hh:mm
6004	time to	hh:mm
6005	noise optimization wind speed	
6006	delay	min
6007	min. wind speed	m/s
6008	max. wind speed	m/s
6009	noise optimization wind direction	
6010	delay	min
6011	angle 1. sector	°
6012	range 1. sector	°
6013	angle 2. sector	°
6014	range 2. sector	°
6015	angle 3. sector	°
6016	range 3. sector	°
6017	combination	
6100	shadow shut off	
6101	Shut off intensity	%
6102	filter on	s
6103	filter off	s

3.3 The Loc/Sens node

This node contains the data of the sensor units group.

3.3.1 The Loc/Sens/Plant<N> node

This node contains the data of one sensor unit (e.g. medium-voltage acquisition).

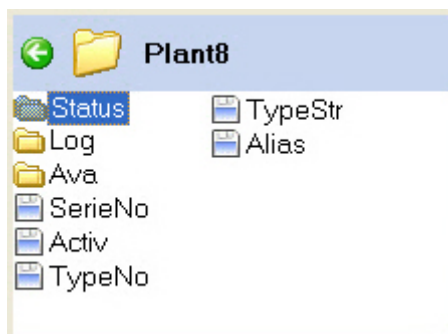


Fig. 15: The *Loc/Sens/Plant<N>* node

Node/Item	Description
Status	Availability data for this sensor unit
Log	Report data for this sensor unit
Ava	Availability data for this sensor unit
SerieNo	Serial number assigned by ENERCON
Active	WEC is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	Identification selectable by customer

3.3.2 The Loc/Sens/Plant<N>/Log/Mspgerf1/<Period> node

This node contains the report data of the medium-voltage acquisition data logger for different periods of time.

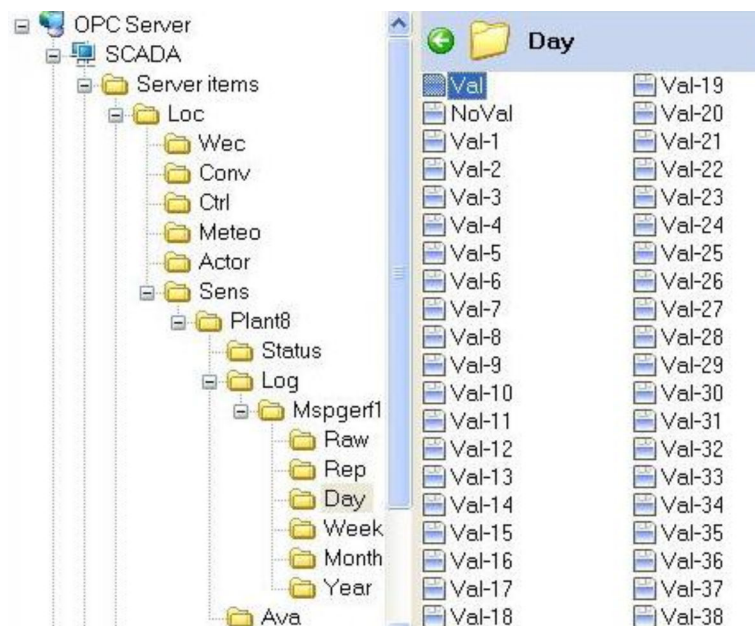


Fig. 16: The *Loc/Sens/Plant<N>/Log/Mspgerf1/<Period>* node

The *NoVal* item shows the number of data records. Depending on the node that contains the report data, the values of the *Val-<X>* item shown in an array (ID 2) have the meaning explained below.

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average voltage U12 [V]	1	1	1	1	1	1
Maximum voltage U12 [V]	2	2	2	2	2	2
Minimum voltage U12 [V]	3	3	3	3	3	3
Average voltage U23 [V]	4	4	4	4	4	4
Maximum voltage U23 [V]	5	5	5	5	5	5
Minimum voltage U23 [V]	6	6	6	6	6	6
Average voltage U31 [V]	7	7	7	7	7	7
Maximum voltage U31 [V]	8	8	8	8	8	8
Minimum voltage U31 [V]	9	9	9	9	9	9
Average frequency [Hz]	10	10	10	10	10	10
Maximum frequency [Hz]	11	11	11	11	11	11
Minimum frequency [Hz]	12	12	12	12	12	12

3.4 The Loc/Conv node

This node contains the data of the converter units group, e.g. solar inverters.

3.4.1 The Loc/Conv/Plant<N> node

This node contains the data of one converter unit, e.g. a solar inverter.

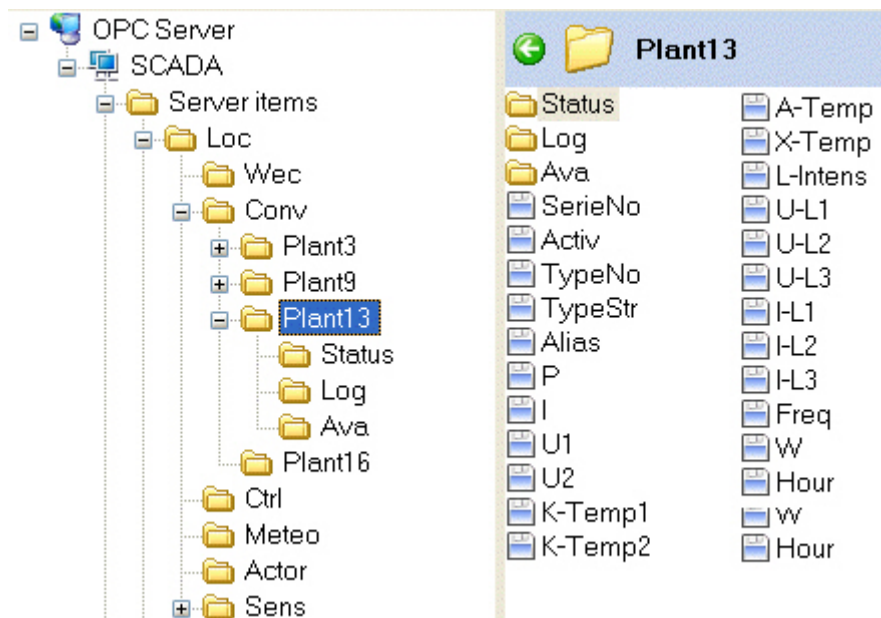


Fig. 17: The Loc/Conv/Plant<N> node

Node/Item	Description
SerieNo	Serial number assigned by ENERCON
Active	WEC is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	Identification selectable by customer
P	Power [kW]
I	Current [A]
U1	Voltage, Level 1 [V]
U2	Voltage, Level 2 [V]
K-Temp1	-----
K-Temp2	-----
A-Temp	Ambient temperature [°C]
X-Temp	Freely selectable temperature [°C]
L-Intens	-----
U-L1	Low voltage U12 [V]
U-L2	Low voltage U23 [V]
U-L3	Low voltage U31 [V]

Node/Item	Description
I-L1	Current feed-in L1 [A]
I-L2	Current feed-in L2 [A]
I-L3	Current feed-in L3 [A]
Freq	Grid frequency [Hz]
W	Energy [kWh]
Hour	Operating hours counter [h]

3.4.2 The Loc/Conv/Plant<N>/Log/Solar1/<Period> node

This node contains the report data of the solar power system data logger for different periods of time.

The numerical value shows the index in the array. If there is no numerical value, this measured value does not exist in that array.

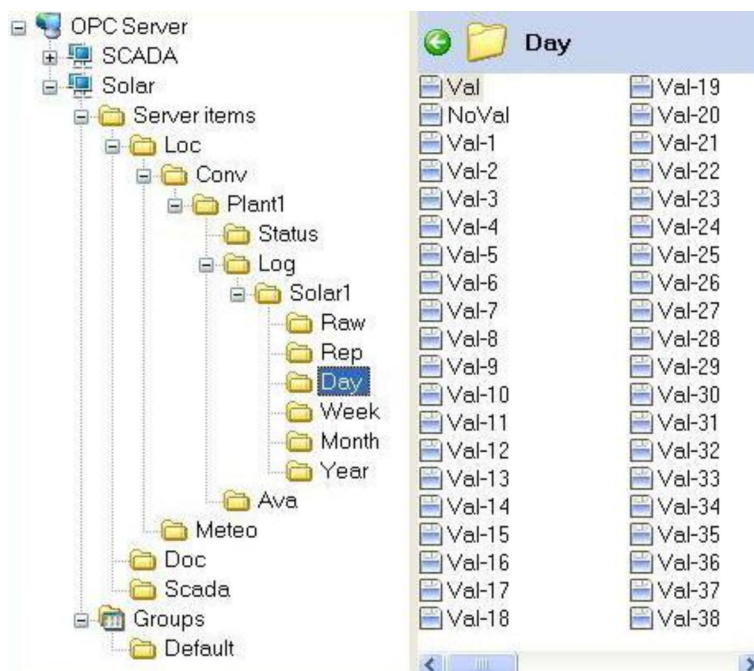


Fig. 18: The *Loc/Conv/Plant<N>/Log/Solar1/<Period>* node

Description	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average power [kW]	1	1	1	1	1	1
Maximum power [kW]	2	2	2	2	2	2
Minimum power [kW]	3	3	3	3	3	3
Average current [A]	4	4	4	4	4	4
Maximum current [A]	5	5	5	5	5	5
Minimum current [A]	6	6	6	6	6	6
Average voltage, Level 1 [V]	7	7	7	7	7	7
Maximum voltage, Level 1 [V]	8	8	8	8	8	8

Description	<i>Raw</i>	<i>Rep</i>	<i>Day</i>	<i>Week</i>	<i>Month</i>	<i>Year</i>
Minimum voltage, Level 1 [V]	9	9	9	9	9	9
Average voltage, Level 2 [V]	10	10	10	10	10	10
Maximum voltage, Level 2 [V]	11	11	11	11	11	11
Minimum voltage, Level 2 [V]	12	12	12	12	12	12
Average mrSolIntens	13	13	13	13	13	13
Maximum mrSolIntens	14	14	14	14	14	14
Minimum IrSolIntens	15	15	15	15	15	15
Average voltage, Level U12 [V]	16	16	16	16	16	16
Maximum voltage, Level U12 [V]	17	17	17	17	17	17
Minimum voltage, Level U12 [V]	18	18	18	18	18	18
Average voltage, Level U23 [V]	19	19	19	19	19	19
Maximum voltage, Level U23 [V]	20	20	20	20	20	20
Minimum voltage, Level U23 [V]	21	21	21	21	21	21
Average voltage, Level U31 [V]	22	22	22	22	22	22
Maximum voltage, Level U31 [V]	23	23	23	23	23	23
Minimum voltage, Level U31 [V]	24	24	24	24	24	24
Average current L1 [A]	25	25	25	25	25	25
Maximum current L1 [A]	26	26	26	26	26	26
Minimum current L1 [A]	27	27	27	27	27	27
Average current L2 [A]	28	28	28	28	28	28
Maximum current L2 [A]	29	29	29	29	29	29
Minimum current L2 [A]	30	30	30	30	30	30
Average current L3 [A]	31	31	31	31	31	31
Maximum current L3 [A]	32	32	32	32	32	32
Minimum current L3 [A]	33	33	33	33	33	33
Average frequency [Hz]	34	34	34	34	34	34
Maximum frequency [Hz]	35	35	35	35	35	35
Minimum frequency [Hz]	36	36	36	36	36	36
Energy produced [kWh]	37	37	37	37	37	37
Operating hours [h]	38	38	38	38	38	38

3.5 The Loc/Ctrl node

This node contains the data of the control units group.

3.5.1 RTU

This node contains the data of the ENERCON SCADA RTU.

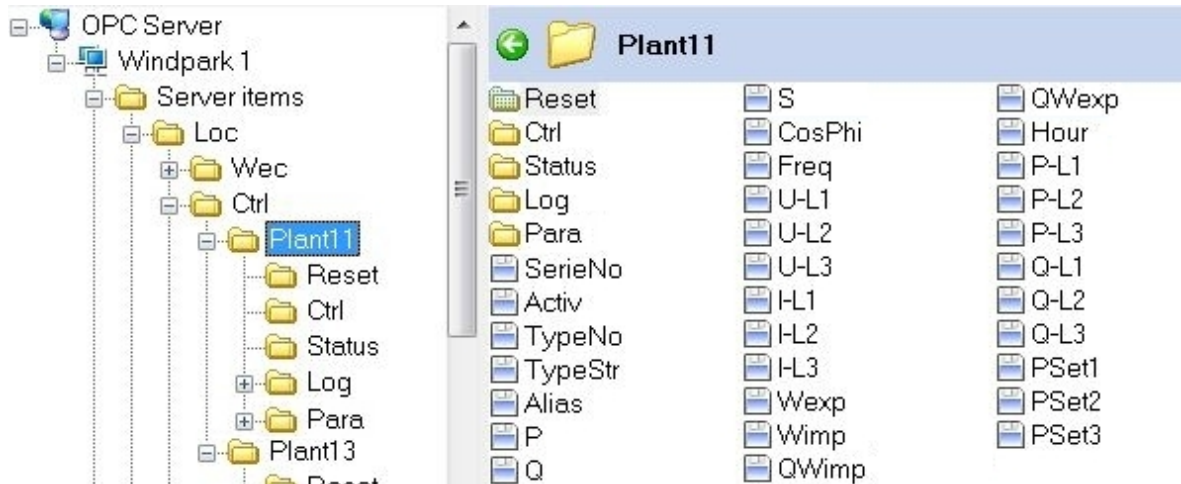


Fig. 19: The *Loc/Ctrl/Plant<N>* for RTU node

Node/Item	Description
Reset	Control system reset (WEC)
Ctrl	Items for RTU control
Status	RTU status data
Log	RTU data logger
Para	RTU parameters
SerieNo	Serial number assigned by ENERCON
Active	WEC is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	Identification selectable by customer
I	Power [kW]
Q	Reactive power [kvar]
CosPhi	Power factor Cos φ
Freq	Frequency [Hz]
U-L1	Voltage U12 [V]
U-L2	Voltage U23 [V]
U-L3	Voltage U31 [V]
I-L1	Current L1 [A]
I-L2	Current L2 [A]

Node/Item	Description
I-L3	Current L3 [A]
Wexp	Exported energy [kWh]
Wimp	Imported energy [kWh]
QWimp	Reactive power during energy import [kvar]
QWexp	Reactive power during energy export [kvar]
Hour	Operating hours [h]
P-L1	Active power L1 [kW]
P-L2	Active power L2 [kW]
P-L3	Active power L3 [kW]
Q-L1	Reactive power L1 [kvar]
Q-L2	Reactive power L2 [kvar]
Q-L3	Reactive power L3 [kvar]
PSet1	Active power setpoint 1 [kW]
PSet2	Active power setpoint 2 [kW]
PSet3	Active power setpoint 3 [kW]

3.5.1.1 The Loc/Ctrl/Plant<N>/Log/Rtu1/<Period> node

This node contains the report data of the RTU data logger for different periods of time.

The numerical value shows the index in the array.

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average power [kW]	1	1	1	1	1	1
Maximum power [kW]	2	2	2	2	2	2
Minimum power [kW]	3	3	3	3	3	3
Average reactive power [kvar]	4	4	4	4	4	4
Maximum reactive power [kvar]	5	5	5	5	5	5
Minimum reactive power [kvar]	6	6	6	6	6	6
Average apparent power [VA]	7	7	7	7	7	7
Maximum apparent power [VA]	8	8	8	8	8	8
Minimum apparent power [VA]	9	9	9	9	9	9
Average cos ϕ	10	10	10	10	10	10
Maximum cos ϕ	11	11	11	11	11	11
Minimum cos ϕ	12	12	12	12	12	12
Average frequency [Hz]	13	13	13	13	13	13
Maximum frequency [Hz]	14	14	14	14	14	14
Minimum frequency [Hz]	15	15	15	15	15	15
Average voltage U12 [V]	16	16	16	16	16	16
Maximum voltage U12 [V]	17	17	17	17	17	17
Minimum voltage U12 [V]	18	18	18	18	18	18
Average voltage U23 [V]	19	19	19	19	19	19
Maximum voltage U23 [V]	20	20	20	20	20	20
Minimum voltage U23 [V]	21	21	21	21	21	21
Average voltage U31 [V]	22	22	22	22	22	22
Maximum voltage U31 [V]	23	23	23	23	23	23
Minimum voltage U31 [V]	24	24	24	24	24	24
Average current 1 L1 [A]	25	25	25	25	25	25
Maximum current 1 L1 [A]	26	26	26	26	26	26
Minimum current 1 L1 [A]	27	27	27	27	27	27
Average current 1 L2 [A]	28	28	28	28	28	28
Maximum current 1 L2 [A]	29	29	29	29	29	29

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Minimum current 1 L2 [A]	30	30	30	30	30	30
Average current 1 L3 [A]	31	31	31	31	31	31
Maximum current 1 L3 [A]	32	32	32	32	32	32
Minimum current 1 L3 [A]	33	33	33	33	33	33
Energy meter, production [kWh]	34	34	34	34	34	34
Energy meter, consumption	35	35	35	35	35	35
Reactive power meter (inductive)	36	36	36	36	36	36
Reactive power meter (capacitive)	37	37	37	37	37	37
Operating hours [h]	38	38	38	38	38	38
Average active power setpoint 1 [kW]	39	39	39	39	39	39
Average active power setpoint 2 [kW]	40	40	40	40	40	40
Average active power setpoint 3 [kW]	41	41	41	41	41	41

3.5.1.2 The Loc/Ctrl/Plant<N>/Log/Rtu15/<Period> node (optional)

This optional node contains the report data of the RTU data logger as 15-minute data for different historical periods of time. The numerical value shows the index in the array.

Description	Interval	
	Raw	Rep
Number of sampling values	0	0
RTU: Average power [kW]	1	1
RTU: Average reactive power [kvar]	2	2
RTU: Average apparent power [VA]	3	3
RTU: Energy meter, production [kWh]	4	4
RTU: Energy meter, consumption	5	5
RTU: Reactive power meter, inductive	6	6
RTU: Reactive power meter, capacitive	7	7
RTU: Operating hours [h]	8	8

3.5.1.3 The Loc/Ctrl/Plant<N>/Log/Rtuphase1/<Period> node

This node contains the report data of the RTU data logger for different periods of time.

The numerical value shows the index in the array.

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average power L1	1	1	1	1	1	1
Maximum power L1	2	2	2	2	2	2
Minimum power L1	3	3	3	3	3	3
Average power L2	4	4	4	4	4	4
Maximum power L2	5	5	5	5	5	5
Minimum power L2	6	6	6	6	6	6
Average power L3	7	7	7	7	7	7
Maximum power L3	8	8	8	8	8	8
Minimum power L3	9	9	9	9	9	9
Average reactive power L1	10	10	10	10	10	10
Maximum reactive power L1	11	11	11	11	11	11
Minimum reactive power L1	12	12	12	12	12	12
Average reactive power L2	13	13	13	13	13	13
Maximum reactive power L2	14	14	14	14	14	14
Minimum reactive power L2	15	15	15	15	15	15
Average reactive power L3	16	16	16	16	16	16
Maximum reactive power L3	17	17	17	17	17	17
Minimum reactive power L3	18	18	18	18	18	18

3.5.1.4 The Loc/Ctrl/Plant<N>/Para for RTU node

The following parameters are available for the ENERCON SCADA RTU:

Item	Description	Unit
1000	Rated Power	kW
1001	Rated Reactive power	Kvar
1002	Rated Voltage	V
1003	PCC Capacity	kW
1004	Rated Frequency	Hz
1010	Serial Number	
1011	RTU Type	
1050	Start Up Voltage	%
1051	Start Up Time	s
1060	Start Up Voltage	%
1061	Start Up Time	s
3000	Scale V Primary	V
3001	Scale V Secondary	V
3002	Scale I Primary	A
3003	Scale I Secondary	A
3012	Measurement Mode	
4000	P Input Bit1	%
4001	P Input Bit2	%
4002	P Input Bit3	%
4003	P Input Bit4	%
4004	P Input Bit5	%
4005	P Input Bit6	%
4006	P Input Bit7	%
4007	P Input Bit8	%
4010	PF Input Bit1	
4011	PF Input Bit2	
4012	PF Input Bit3	
4013	PF Input Bit4	
4014	PF Input Bit5	
4015	PF Input Bit6	
4016	PF Input Bit7	
4017	PF Input Bit8	

Item	Description	Unit
4020	Q Input Bit1	%
4021	Q Input Bit2	%
4022	Q Input Bit3	%
4023	Q Input Bit4	%
4024	Q Input Bit5	%
4025	Q Input Bit6	%
4026	Q Input Bit7	%
4027	Q Input Bit8	%
4030	U Input Bit1	%
4031	U Input Bit2	%
4032	U Input Bit3	%
4033	U Input Bit4	%
4034	U Input Bit5	%
4035	U Input Bit6	%
4036	U Input Bit7	%
4037	U Input Bit8	%
4040	P Exception	%
4041	PF Exception	
4042	Q Exception	%
4050	Signal Fault Behav.	
4060	WF Stop Mode	
4104	P Min Input	%
4105	P Max Input	%
4114	PF Min Input	
4115	PF Max Input	
4124	Q Min Input	%
4125	Q Max Input	%
4134	U Min Input	%
4135	U Max Input	%
4200	Analog Out 5 Type	
4201	Analog Out 5 Min.	
4202	Analog Out 5 Max.	
4203	Analog Out 5 Signal	
4210	Analog Out 6 Type	

Item	Description	Unit
4211	Analog Out 6 Min.	
4212	Analog Out 6 Max.	
4213	Analog Out 6 Signal	
4220	Analog Out 7 Type	
4221	Analog Out 7 Min.	
4222	Analog Out 7 Max.	
4223	Analog Out 7 Signal	
4230	Analog Out 8 Type	
4231	Analog Out 8 Min.	
4232	Analog Out 8 Max.	
4233	Analog Out 8 Signal	
4240	Analog Out 9 Type	
4241	Analog Out 9 Min.	
4242	Analog Out 9 Max.	
4243	Analog Out 9 Signal	
4250	Analog Out 10 Type	
4251	Analog Out 10 Min.	
4252	Analog Out 10 Max.	
4253	Analog Out 10 Signal	
4260	Analog Out 11 Type	
4261	Analog Out 11 Min.	
4262	Analog Out 11 Max.	
4263	Analog Out 11 Signal	
4270	Analog Out 12 Type	
4271	Analog Out 12 Min.	
4272	Analog Out 12 Max.	
4273	Analog Out 12 Signal	
4300	P Set Point Src.	
4301	PF Set Point Src.	
4302	Q Set Point Src.	
4303	U Set Point Src.	
4304	Freq. Res. Cap. Src.	
4310	P Control Mode	
4311	Q Control Mode	

Item	Description	Unit
4320	P Set Point Src. – Source 2	
4350	P Default Set Point	%
4351	PF Default Set Point	
4352	Q Default Set Point	%
4353	U Default Set Point	%
4354	Freq. Res. Cap. Def.	%
4360	P Display Set Point	%
4361	PF Display Set Point	
4362	Q Display Set Point	%
4363	U Display Set Point	%
4364	Freq. Res. Cap. Disp	%
6000	Crv Cos Phi(P) Nodes	
6001	Curve Q(U) Nodes	
6002	Crv Cos Phi(U) Nodes	
6003	Curve Q(P) Nodes	
6010	Crv Cos Phi(P) Timer	s
6011	CurveQ(U) Timer	s
6012	Crv Cos Phi(U) Timer	s
6013	Curve Q(P) Timer	s
6020	Curve Cos Phi Min.	
6021	Curve Cos Phi Max.	
6031	CurveQ(U) Dead band	%
6032	Crv Cos Phi(U) Dead band	%
6100	Curve Cos Phi(P) X1	%
6101	Curve Cos Phi(P) Y1	%
6102	Curve Cos Phi(P) X2	%
6103	Curve Cos Phi(P) Y2	%
6104	Curve Cos Phi(P) X3	%
6105	Curve Cos Phi(P) Y3	%
6106	Curve Cos Phi(P) X4	%
6107	Curve Cos Phi(P) Y4	%
6108	Curve Cos Phi(P) X5	%
6109	Curve Cos Phi(P) Y5	%
6110	Curve Cos Phi(P) X6	%

Item	Description	Unit
6111	Curve Cos Phi(P) Y6	%
6112	Curve Cos Phi(P) X7	%
6113	Curve Cos Phi(P) Y7	%
6114	Curve Cos Phi(P) X8	%
6115	Curve Cos Phi(P) Y8	%
6116	Curve Cos Phi(P) X9	%
6117	Curve Cos Phi(P) Y9	%
6118	Curve Cos Phi(P) X10	%
6119	Curve Cos Phi(P) Y10	%
6120	Curve Cos Phi(P) X11	%
6121	Curve Cos Phi(P) Y11	%
6122	Curve Cos Phi(P) X12	%
6123	Curve Cos Phi(P) Y12	%
6150	CurveQ(U) X1	%
6151	CurveQ(U) Y1	%
6152	CurveQ(U) X2	%
6153	CurveQ(U) Y2	%
6154	CurveQ(U) X3	%
6155	CurveQ(U) Y3	%
6156	CurveQ(U) X4	%
6157	CurveQ(U) Y4	%
6158	CurveQ(U) X5	%
6159	CurveQ(U) Y5	%
6160	CurveQ(U) X6	%
6161	CurveQ(U) Y6	%
6162	CurveQ(U) X7	%
6163	CurveQ(U) Y7	%
6164	CurveQ(U) X8	%
6165	CurveQ(U) Y8	%
6166	CurveQ(U) X9	%
6167	CurveQ(U) Y9	%
6168	CurveQ(U) X10	%
6169	CurveQ(U) Y10	%
6170	CurveQ(U) X11	%

Item	Description	Unit
6171	CurveQ(U) Y11	%
6172	CurveQ(U) X12	%
6173	CurveQ(U) Y12	%
6200	Curve Cos Phi(U) X1	%
6201	Curve Cos Phi(U) Y1	%
6202	Curve Cos Phi(U) X2	%
6203	Curve Cos Phi(U) Y2	%
6204	Curve Cos Phi(U) X3	%
6205	Curve Cos Phi(U) Y3	%
6206	Curve Cos Phi(U) X4	%
6207	Curve Cos Phi(U) Y4	%
6208	Curve Cos Phi(U) X5	%
6209	Curve Cos Phi(U) Y5	%
6210	Curve Cos Phi(U) X6	%
6211	Curve Cos Phi(U) Y6	%
6212	Curve Cos Phi(U) X7	%
6213	Curve Cos Phi(U) Y7	%
6214	Curve Cos Phi(U) X8	%
6215	Curve Cos Phi(U) Y8	%
6216	Curve Cos Phi(U) X9	%
6217	Curve Cos Phi(U) Y9	%
6218	Curve Cos Phi(U) X10	%
6219	Curve Cos Phi(U) Y10	%
6220	Curve Cos Phi(U) X11	%
6221	Curve Cos Phi(U) Y11	%
6222	Curve Cos Phi(U) X12	%
6223	Curve Cos Phi(U) Y12	%
6250	Curve Q(P) X1	%
6251	Curve Q(P) Y1	%
6252	Curve Q(P) X2	%
6253	Curve Q(P) Y2	%
6254	Curve Q(P) X3	%
6255	Curve Q(P) Y3	%
6256	Curve Q(P) X4	%

Item	Description	Unit
6257	Curve Q(P) Y4	%
6258	Curve Q(P) X5	%
6259	Curve Q(P) Y5	%
6260	Curve Q(P) X6	%
6261	Curve Q(P) Y6	%
6262	Curve Q(P) X7	%
6263	Curve Q(P) Y7	%
6264	Curve Q(P) X8	%
6265	Curve Q(P) Y8	%
6266	Curve Q(P) X9	%
6267	Curve Q(P) Y9	%
6268	Curve Q(P) X10	%
6269	Curve Q(P) Y10	%
6270	Curve Q(P) X11	%
6271	Curve Q(P) Y11	%
6272	Curve Q(P) X12	%
6273	Curve Q(P) Y12	%

3.5.2 GDA

This node contains data of the ENERCON SCADA GDA.

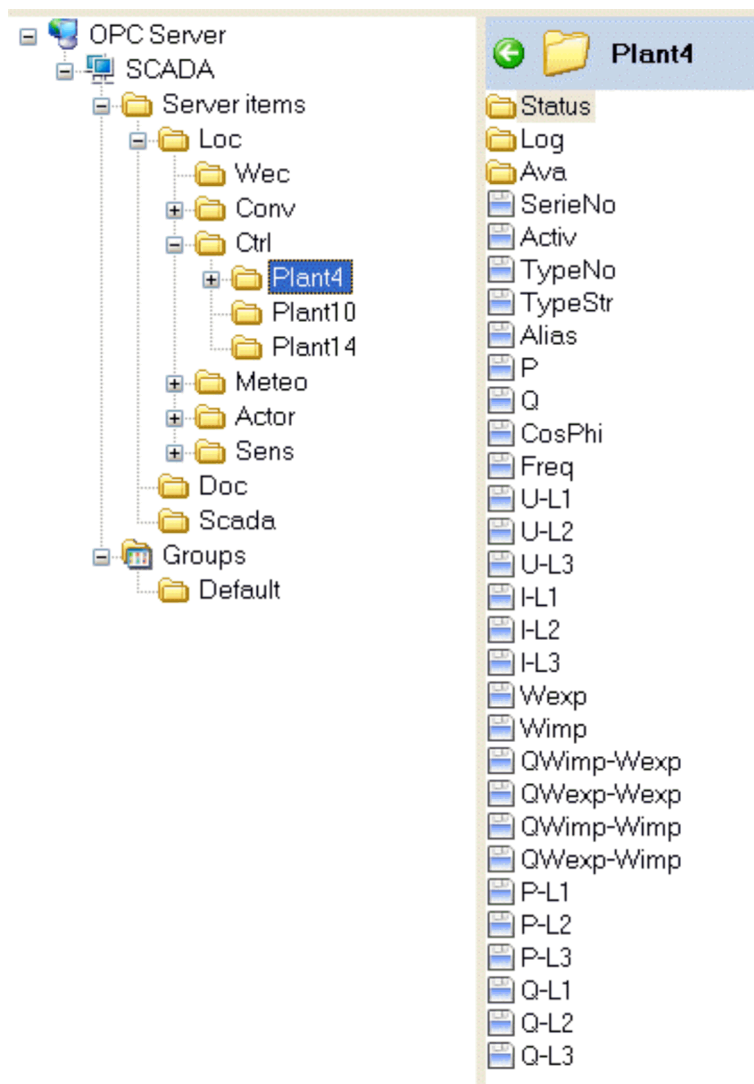


Fig. 20: The *Loc/Ctrl/Plant<N>* for GDA node

Node/Item	Description
SerieNo	Serial number assigned by ENERCON
Active	WEC is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	Identification selectable by customer
P	Power [kW]
Q	Reactive power [kvar]
CosPhi	Power factor cosφ
Freq	Frequency [Hz]
U-L1	Low voltage U12 [V]

Node/Item	Description
U-L2	Low voltage U23 [V]
U-L3	Low voltage U31 [V]
I-L1	Current feed-in L1 [A]
I-L2	Current feed-in L2 [A]
I-L3	Current feed-in L3 [A]
Wexp	Exported energy [kWh]
Wimp	Imported energy [kWh]
QWimp-Wexp	Reactive power import during active power export [kvarh]
QWexp-Wexp	Reactive power export during active power export [kvarh]
QWimp-Wimp	Reactive power import during active power import [kvarh]
QWexp-Wimp	Reactive power export during active power import [kvarh]
P-L1	Power L1 [kW]
P-L2	Power L2 [kW]
P-L3	Power L3 [kW]
Q-L1	Reactive power L1 [kvar]
Q-L2	Reactive power L2 [kvar]
Q-L3	Reactive power L3 [kvar]

3.5.2.1 The Loc/Ctrl/Plant<N>/Log/Farmctr1/<Period> node

This node contains the report data of the GDA data logger for different periods of time.

The numerical value shows the index in the array. If there is no numerical value, this measured value does not exist in that array.

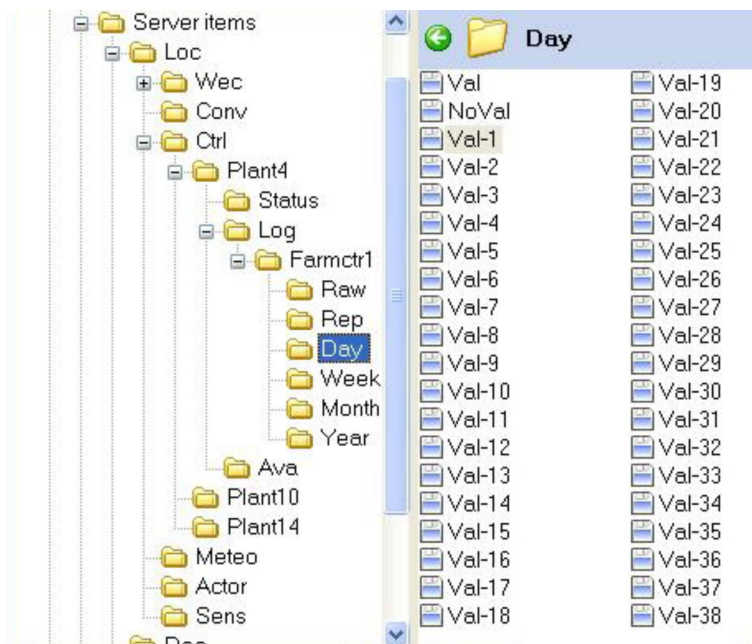


Fig. 21: The *Loc/Ctrl/Plant<N>/Log/Farmctr1/Day* node

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average power [kW]	1	1	1	1	1	1
Maximum power [kW]	2	2	2	2	2	2
Minimum power [kW]	3	3	3	3	3	3
Average reactive power [kvar]	4	4	4	4	4	4
Maximum reactive power [kvar]	5	5	5	5	5	5
Minimum reactive power [kvar]	6	6	6	6	6	6
Average cos ϕ	7	7	7	7	7	7
Maximum cos ϕ	8	8	8	8	8	8
Minimum cos ϕ	9	9	9	9	9	9
Average frequency [Hz]	10	10	10	10	10	10
Maximum frequency [Hz]	11	11	11	11	11	11
Minimum frequency [Hz]	12	12	12	12	12	12
Average voltage U12 [V]	13	13	13	13	13	13
Maximum voltage U12 [V]	14	14	14	14	14	14
Minimum voltage U12 [V]	15	15	15	15	15	15

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Average voltage U23 [V]	16	16	16	16	16	16
Maximum voltage U23 [V]	17	17	17	17	17	17
Minimum voltage U23 [V]	18	18	18	18	18	18
Average voltage U31 [V]	19	19	19	19	19	19
Maximum voltage U31 [V]	20	20	20	20	20	20
Minimum voltage U31 [V]	21	21	21	21	21	21
Average current L1 [A]	22	22	22	22	22	22
Maximum current L1 [A]	23	23	23	23	23	23
Minimum current L1 [A]	24	24	24	24	24	24
Average current L2 [A]	25	25	25	25	25	25
Maximum current L2 [A]	26	26	26	26	26	26
Minimum current L2 [A]	27	27	27	27	27	27
Average current L3 [A]	28	28	28	28	28	28
Maximum current L3 [A]	29	29	29	29	29	29
Minimum current L3 [A]	30	30	30	30	30	30
Exported energy [kWh]	31	31	31	31	31	31
Imported energy [kWh]	32	32	32	32	32	32
Reactive power (inductive) 1 [kvar]	33	33	33	33	33	33
Reactive power (capacitive) 1 [kvar]	34	34	34	34	34	34
Reactive power (inductive) 2 [kvar]	35	35	35	35	35	35
Reactive power (capacitive) 2 [kvar]	36	36	36	36	36	36

3.5.2.2 The Loc/Ctrl/Plant<N>/Log/Farmctr15/<Period> node (optional)

This optional node contains the report data of the ENERCON GDA data logger as 15-minute data for different historical periods of time. The numerical value shows the index in the array.

Description	Interval	
	Raw	Rep
Number of sampling values	0	0
GDA: Average power	1	1
GDA: Average reactive power	2	2
GDA: Energy meter (production)	3	3
GDA: Energy meter (consumption)	4	4
GDA: Reactive power meter (inductive) for active power generation	5	5
GDA: Reactive power meter (capacitive) for active power generation	6	6
GDA: Reactive power meter (inductive) for active power consumption	7	7
GDA: Reactive power meter (capacitive) for active power consumption	8	8

3.5.3 FCU

This node contains data of the ENERCON Farm Control Unit (FCU).

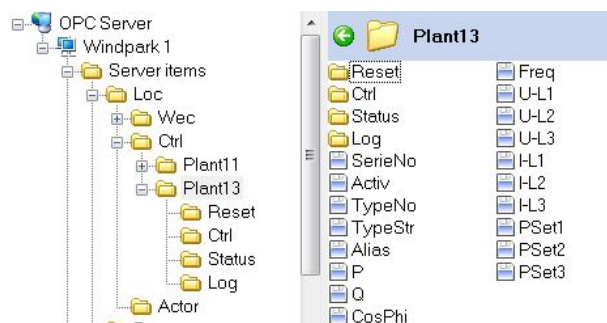


Fig. 22: Loc/Ctrl/Plant<N> node for FCU

Node/Item	Description
Reset	Control system reset (WEC)
Ctrl	Items for FCU control
Status	FCU status data
Log	FCU data logger
SerieNo	Serial number assigned by ENERCON
Active	WEC is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	Identification selectable by customer
I	Power [kW]
Q	Reactive power [kvar]
CosPhi	Power factor Cos ϕ
Freq	Frequency [Hz]
U-L1	Voltage U12 [V]
U-L2	Voltage U23 [V]
U-L3	Voltage U31 [V]
I-L1	Current L1 [A]
I-L2	Current L2 [A]
I-L3	Current L3 [A]
PSet1	Active power setpoint 1 [kW]
PSet2	Active power setpoint 2 [kW]
PSet3	Active power setpoint 3 [kW]

3.5.3.1 The Loc/Ctrl/Plant<N>/Log/Fcuctr1/<Period> node

This node contains the report data of the FCU data logger for different periods of time.

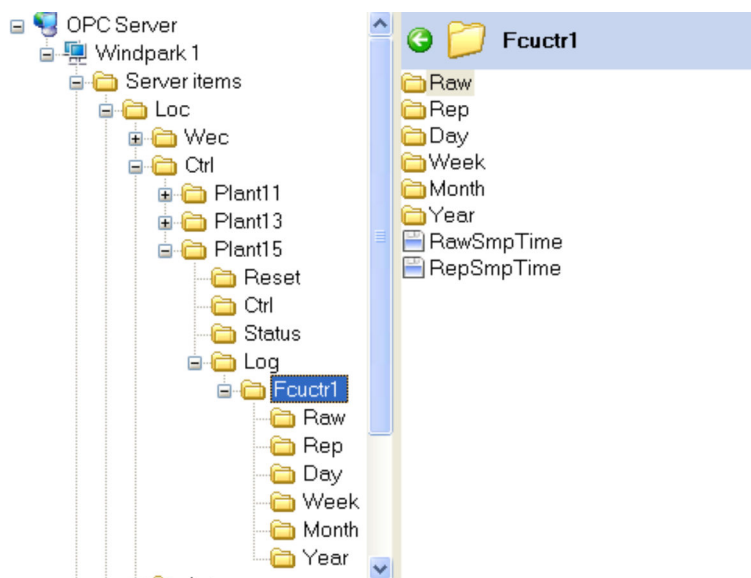


Fig. 23: The Loc/Ctrl/Plant<N>/Log/Fcuctr1/<Period> node

The numerical value shows the index in the array. If there is no numerical value, this measured value does not exist in that array.

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average power [kW]	1	1	1	1	1	1
Maximum power [kW]	2	2	2	2	2	2
Minimum power [kW]	3	3	3	3	3	3
Average reactive power [kvar]	4	4	4	4	4	4
Maximum reactive power [kvar]	5	5	5	5	5	5
Minimum reactive power [kvar]	6	6	6	6	6	6
Average cos φ	7	7	7	7	7	7
Maximum cos φ	8	8	8	8	8	8
Minimum cos φ	9	9	9	9	9	9
Average frequency [Hz]	10	10	10	10	10	10
Maximum frequency [Hz]	11	11	11	11	11	11
Minimum frequency [Hz]	12	12	12	12	12	12
Average voltage U12 [V]	13	13	13	13	13	13
Maximum voltage U12 [V]	14	14	14	14	14	14
Minimum voltage U12 [V]	15	15	15	15	15	15
Average voltage U23 [V]	16	16	16	16	16	16

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Maximum voltage U23 [V]	17	17	17	17	17	17
Minimum voltage U23 [V]	18	18	18	18	18	18
Average voltage U31 [V]	19	19	19	19	19	19
Maximum voltage U31 [V]	20	20	20	20	20	20
Minimum voltage U31 [V]	21	21	21	21	21	21
Average current L1 [A]	22	22	22	22	22	22
Maximum current L1 [A]	23	23	23	23	23	23
Minimum current L1 [A]	24	24	24	24	24	24
Average current L2 [A]	25	25	25	25	25	25
Maximum current L2 [A]	26	26	26	26	26	26
Minimum current L2 [A]	27	27	27	27	27	27
Average current L3 [A]	28	28	28	28	28	28
Maximum current L3 [A]	29	29	29	29	29	29
Minimum current L3 [A]	30	30	30	30	30	30
Average active power setpoint 1 [kW]	31	31	31	31	31	31
Average active power setpoint 2 [kW]	32	32	32	32	32	32
Average active power setpoint 3 [kW]	33	33	33	33	33	33

3.5.4 SCU

This node contains data for the ENERCON SCU.

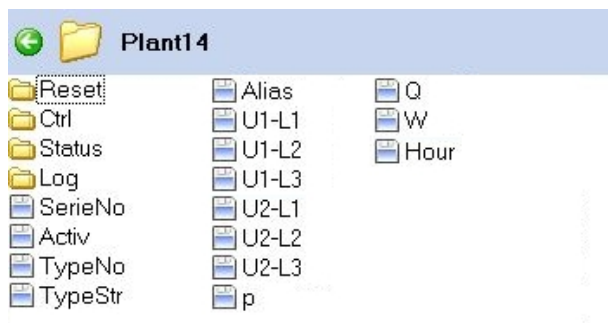


Fig. 24: Loc/Ctrl/Plant<N> node for transmission substation 1

Node/Item	Description
Status	Status data
Log	Report data
Ava	Availability data
SerieNo	Serial number assigned by ENERCON
Active	WEC is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	Identification selectable by customer
U1-L1	Voltage, Level 1 U12 [V]
U1-L2	Voltage, Level 1 U23 [V]
U1-L3	Voltage, Level 1 U31 [V]
U2-L1	Voltage, Level 2 U12 [V]
U2-L2	Voltage, Level 2 U23 [V]
U2-L3	Voltage, Level 2 U31 [V]
P	Power [kW]
Q	Reactive power [kvar]
W	Energy [kWh]
Hour	Operating hours [h]

3.5.4.1 The Loc/Ctrl/Plant<N>/Log/Substat1/<Period> node

This node contains the report data of the SCU data logger for different periods of time.

The numerical value shows the index in the array. If there is no numerical value, this measured value does not exist in that array.

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average voltage 1 U12 [V]	1	1	1	1	1	1
Average voltage 1 U23 [V]	2	2	2	2	2	2
Average voltage 1 U31 [V]	3	3	3	3	3	3
Average voltage 2 U12 [V]	4	4	4	4	4	4
Average voltage 2 U23 [V]	5	5	5	5	5	5
Average voltage 2 U31 [V]	6	6	6	6	6	6
Average power [kW]	7	7	7	7	7	7
Average reactive power [kvar]	8	8	8	8	8	8

3.5.5 CLC

This node contains data of the ENERCON CLC.

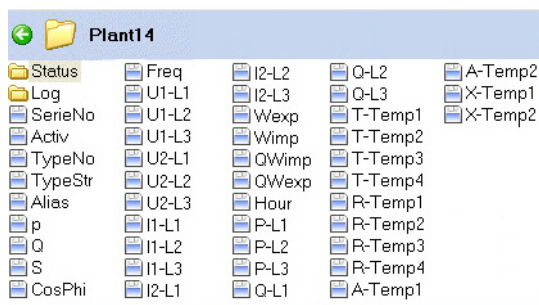


Fig. 25: Loc/Ctrl/Plant<N> node for transmission substation 2

Node/Item	Description
SerieNo	Serial number assigned by ENERCON
Active	WEC is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	Identification selectable by customer
P	Power [kW]
Q	Reactive power [kvar]
S	Apparent power [VA]
CosPhi	Power factor Cosφ
Freq	Frequency [Hz]
U1-L1	Voltage, Level 1 U12 [V]
U1-L2	Voltage, Level 1 U23 [V]
U1-L3	Voltage, Level 1 U31 [V]
U2-L1	Voltage, Level 2 U12 [V]
U2-L2	Voltage, Level 2 U23 [V]
U2-L3	Voltage, Level 2 U31 [V]
I1-L1	Current, Level 1 L1 [A]
I1-L2	Current, Level 1 L2 [A]
I1-L3	Current, Level 1 L3 [A]
I2-L1	Current, Level 2 L1 [A]
I2-L2	Current, Level 2 L2 [A]
I2-L3	Current, Level 2 L3 [A]
Wexp	Exported energy [kWh]
Wimp	Imported energy [kWh]
QWimp	Reactive power import [kVarh]

Node/Item	Description
QWexp	Reactive power export [kVarh]
Hour	Operating hours [h]
P-L1	Power L1 [kWh]
P-L2	Power L2 [kWh]
P-L3	Power L3 [kWh]
Q-L1	Reactive power L1 [kvar]
Q-L2	Reactive power L2 [kvar]
Q-L3	Reactive power L3 [kvar]
T-Temp1*	Transformer temperature 1 [°C]
T-Temp2*	Transformer temperature 2 [°C]
T-Temp3*	Transformer temperature 3 [°C]
T-Temp4*	Transformer temperature 4 [°C]
R-Temp1*	Room temperature 1 [°C]
R-Temp2*	Room temperature 2 [°C]
R-Temp3*	Room temperature 3 [°C]
R-Temp4*	Room temperature 4 [°C]
A-Temp1*	Ambient temperature 1 [°C]
A-Temp2 *	Ambient temperature 2 [°C]
X-Temp1*	Temperature [°C] (Dummy)
X-Temp2*	Temperature [°C] (Dummy)

The positions of temperature sensors need to be defined for each specific project.

3.5.5.1 The Loc/Ctrl/Plant<N>/Log/Substat2/<Period> node

This node contains the report data of the SCU data logger for different periods of time.

The numerical value shows the index in the array. If there is no numerical value, this measured value does not exist in that array.

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average power [kW]	1	1	1	1	1	1
Maximum power [kW]	2	2	2	2	2	2
Minimum power [kW]	3	3	3	3	3	3
Average reactive power [kvar]	4	4	4	4	4	4
Maximum reactive power [kvar]	5	5	5	5	5	5
Minimum reactive power [kvar]	6	6	6	6	6	6
Average apparent power [kVA]	7	7	7	7	7	7
Maximum apparent power [kVA]	8	8	8	8	8	8
Minimum apparent power [kVA]	9	9	9	9	9	9
Average cos ϕ	10	10	10	10	10	10
Maximum cos ϕ	11	11	11	11	11	11
Minimum cos ϕ	12	12	12	12	12	12
Average frequency [Hz]	13	13	13	13	13	13
Maximum frequency [Hz]	14	14	14	14	14	14
Minimum frequency [Hz]	15	15	15	15	15	15
Average voltage 1 L1 [V]	16	16	16	16	16	16
Maximum voltage 1 L1 [V]	17	17	17	17	17	17
Minimum voltage 1 L1 [V]	18	18	18	18	18	18
Average voltage 1 L2 [V]	19	19	19	19	19	19
Maximum voltage 1 L2 [V]	20	20	20	20	20	20
Minimum voltage 1 L2 [V]	21	21	21	21	21	21
Average voltage 1 L3 [V]	22	22	22	22	22	22
Maximum voltage 1 L3 [V]	23	23	23	23	23	23
Minimum voltage 1 L3 [V]	24	24	24	24	24	24
Average voltage 2 L1 [V]	25	25	25	25	25	25
Maximum voltage 2 L1 [V]	26	26	26	26	26	26
Minimum voltage 2 L1 [V]	27	27	27	27	27	27
Average voltage 2 L2 [V]	28	28	28	28	28	28
Maximum voltage 2 L2 [V]	29	29	29	29	29	29

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Minimum voltage 2 L2 [V]	30	30	30	30	30	30
Average voltage 2 L3 [V]	31	31	31	31	31	31
Maximum voltage 2 L3 [V]	32	32	32	32	32	32
Minimum voltage 2 L3 [V]	33	33	33	33	33	33
Average current 1 L1 [A]	34	34	34	34	34	34
Maximum current 1 L1 [A]	35	35	35	35	35	35
Minimum current 1 L1 [A]	36	36	36	36	36	36
Average current 1 L2 [A]	37	37	37	37	37	37
Maximum current 1 L2 [A]	38	38	38	38	38	38
Minimum current 1 L2 [A]	39	39	39	39	39	39
Average current 1 L3 [A]	40	40	40	40	40	40
Maximum current 1 L3 [A]	41	41	41	41	41	41
Minimum current 1 L3 [A]	42	42	42	42	42	42
Average current 2 L1 [A]	43	43	43	43	43	43
Maximum current 2 L1 [A]	44	44	44	44	44	44
Minimum current 2 L1 [A]	45	45	45	45	45	45
Average current 2 L2 [A]	46	46	46	46	46	46
Maximum current 2 L2 [A]	47	47	47	47	47	47
Minimum current 2 L2 [A]	48	48	48	48	48	48
Average current 2 L3 [A]	49	49	49	49	49	49
Maximum current 2 L3 [A]	50	50	50	50	50	50
Minimum current 2 L3 [A]	51	51	51	51	51	51
Exported energy [kWh]	52	52	52	52	52	52
Imported energy [kWh]	53	53	53	53	53	53
Reactive power (inductive) [kvar]	54	54	54	54	54	54
Reactive power (capacitive) [kvar]	55	55	55	55	55	55
Operating hours [h]	56	56	56	56	56	56

3.5.5.2 The Loc/Ctrl/Plant<N>/Log/SubTemp2/<Period> node

This node contains the report data of the SCU data logger for different periods of time.

The numerical value shows the index in the array. If there is no numerical value, this measured value does not exist in that array.

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average temperature 1 [°C]	1	1	1	1	1	1
Maximum temperature 1 [°C]	2	2	2	2	2	2
Minimum temperature 1 [°C]	3	3	3	3	3	3
Average temperature 2 [°C]	4	4	4	4	4	4
Maximum temperature 2 [°C]	5	5	5	5	5	5
Minimum temperature 2 [°C]	6	6	6	6	6	6
Average temperature 3 [°C]	7	7	7	7	7	7
Maximum temperature 3 [°C]	8	8	8	8	8	8
Minimum temperature 3 [°C]	9	9	9	9	9	9
Average temperature 4 [°C]	10	10	10	10	10	10
Maximum temperature 4 [°C]	11	11	11	11	11	11
Minimum temperature 4 [°C]	12	12	12	12	12	12
Average temperature 5 [°C]	13	13	13	13	13	13
Maximum temperature 5 [°C]	14	14	14	14	14	14
Minimum temperature 5 [°C]	15	15	15	15	15	15
Average temperature 6 [°C]	16	16	16	16	16	16
Maximum temperature 6 [°C]	17	17	17	17	17	17
Minimum temperature 6 [°C]	18	18	18	18	18	18
Average temperature 7 [°C]	19	19	19	19	19	19
Maximum temperature 7 [°C]	20	20	20	20	20	20
Minimum temperature 7 [°C]	21	21	21	21	21	21
Average temperature 8 [°C]	22	22	22	22	22	22
Maximum temperature 8 [°C]	23	23	23	23	23	23
Minimum temperature 8 [°C]	24	24	24	24	24	24
Average temperature 9 [°C]	25	25	25	25	25	25
Maximum temperature 9 [°C]	26	26	26	26	26	26
Minimum temperature 9 [°C]	27	27	27	27	27	27
Average temperature 10 [°C]	28	28	28	28	28	28
Maximum temperature 10 [°C]	29	29	29	29	29	29

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Minimum temperature 10 [°C]	30	30	30	30	30	30
Average temperature 11 [°C]	31	31	31	31	31	31
Maximum temperature 11 [°C]	32	32	32	32	32	32
Minimum temperature 11 [°C]	33	33	33	33	33	33
Average temperature 12 [°C]	34	34	34	34	34	34
Maximum temperature 12 [°C]	35	35	35	35	35	35
Minimum temperature 12 [°C]	36	36	36	36	36	36

3.5.5.3 The Loc/Ctrl/Plant<N>/Log/Subphase2/<Period> node

This node contains the report data of the SCU data logger for different periods of time.

The numerical value shows the index in the array. If there is no numerical value, this measured value does not exist in that array.

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average power L1 [kW]	1	1	1	1	1	1
Maximum power L1 [kW]	2	2	2	2	2	2
Minimum power L1 [kW]	3	3	3	3	3	3
Average power L2 [kW]	4	4	4	4	4	4
Maximum power L2 [kW]	5	5	5	5	5	5
Minimum power L2 [kW]	6	6	6	6	6	6
Average power L3 [kW]	7	7	7	7	7	7
Maximum power L3 [kW]	8	8	8	8	8	8
Minimum power L3 [kW]	9	9	9	9	9	9
Average reactive power L1 [kvar]	10	10	10	10	10	10
Maximum reactive power L1 [kvar]	11	11	11	11	11	11
Minimum reactive power L1 [kvar]	12	12	12	12	12	12
Average reactive power L2 [kvar]	13	13	13	13	13	13
Maximum reactive power L2 [kvar]	14	14	14	14	14	14
Minimum reactive power L2 [kvar]	15	15	15	15	15	15
Average reactive power L3 [kvar]	16	16	16	16	16	16
Maximum reactive power L3 [kvar]	17	17	17	17	17	17
Minimum reactive power L3 [kvar]	18	18	18	18	18	18

3.6 The Loc/Meteo node

This node contains the data of the meteorological units group. One or several ENERCON METEO devices may be installed.

3.6.1 The Loc/Meteo/Plant<N> node

This node contains the ENERCON METEO data.

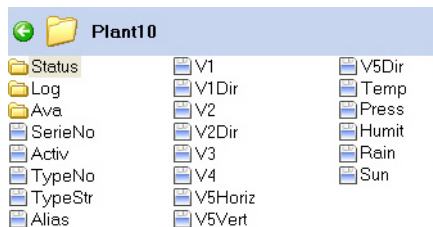


Fig. 26: The *Loc/Meteo/Plant<N>* node

Node/Item	Description
SerieNo	Serial number assigned by ENERCON
Active	WEC is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	Identification selectable by customer
V1	Wind speed at height 1 [m/s]
V1Dir	Wind direction at height 1 [°]
V2	Wind speed at height 2 [m/s]
V2Dir	Wind direction at height 2 [°]
V3	Wind speed at height 3 [m/s]
V4	Wind speed at height 4 [m/s]
V5Horiz	Wind speed (horizontal) at height 5 [m/s]
V5Vert	Wind speed (vertical) at height 5 [m/s]
V5Dir	Wind direction at height 5 [°]
Temp	Outside temperature [°C]
Press	Atmospheric pressure [mbar]
Humit	Air humidity [%]
Rain	Precipitation
Sun	Solar radiation [W/m²]

3.6.2 The Loc/Meteo/Plant<N>/Log/Wind1/<Period> node

This node contains the report data of the ENERCON METEO wind measurement data logger for different periods of time. The numerical value shows the index in the array. If there is no numerical value, this measured value does not exist in that array.

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Number of sampling values	0	0	0	0	0	0
Average wind speed 1 [m/s]	1	1	1	1	1	1
Maximum wind speed 1 [m/s]	2	2	2	2	2	2
Minimum wind speed 1 [m/s]	3	3	3			
Wind direction 1 [°]	4	4	4			
Average wind speed 2 [m/s]	5	5	5	3	3	3
Maximum wind speed 2 [m/s]	6	6	6	4	4	4
Minimum wind speed 2 [m/s]	7	7	7			
Wind direction 2 [°]	8	8	8			
Average X wind direction 1 [°]	9	9	9			
Average Y wind direction 1 [°]	10	10	10			
Average X wind direction 2 [°]	11	11	11			
Average Y wind direction 2 [°]	12	12	12			
Average wind speed 3 [m/s]	13	13	13	5	5	5
Maximum wind speed 3 [m/s]	14	14	14	6	6	6
Minimum wind speed 3 [m/s]	15	15	15			
Average wind speed 4 [m/s]	16	16	16	7	7	7
Maximum wind speed 4 [m/s]	17	17	17	8	8	8
Minimum wind speed 4 [m/s]	18	18	18			
Average horizontal wind speed (ultra-sonic anemometer) [m/s]	19	19	19	9	9	9
Maximum horizontal wind speed (ultra-sonic anemometer) [m/s]	20	20	20	10	10	10
Minimum horizontal wind speed (ultra-sonic anemometer) [m/s]	21	21	21			
Average vertical wind speed (ultrasonic anemometer) [m/s]	22	22	22	11	11	11
Maximum vertical wind speed (ultrasonic anemometer) [m/s]	23	23	23	12	12	12
Minimum vertical wind speed (ultrasonic anemometer) [m/s]	24	24	24			

Description	Interval					
	Raw	Rep	Day	Week	Month	Year
Average X wind direction (ultrasonic anemometer) [°]	25	25	25			
Average Y wind direction (ultrasonic anemometer) [°]	26	26	26			

3.6.3 The Loc/Meteo/Plant<Index>/Log/Weather2/<Period> node

This node contains the report data of the ENERCON METEO weather analysis data logger for different periods of time. If there is no numerical value, this measured value does not exist in that array.

Description	Interval					
	<i>Raw</i>	<i>Rep</i>	<i>Day</i>	<i>Week</i>	<i>Month</i>	<i>Year</i>
Number of sampling values	0	0	0	0	0	0
Ambient temperature [°C]	1	1	1	1	1	1
Atmospheric pressure [mbar]	2	2	2	2	2	2
Air humidity [%]	3	3	3	3	3	3
Precipitation [mm/min]	4	4	4	4	4	4
Solar radiation [W/m²]	5	5	5	5	5	5

3.7 The Loc/Actor node

This node contains the data of the actors group.

3.7.1 PDI

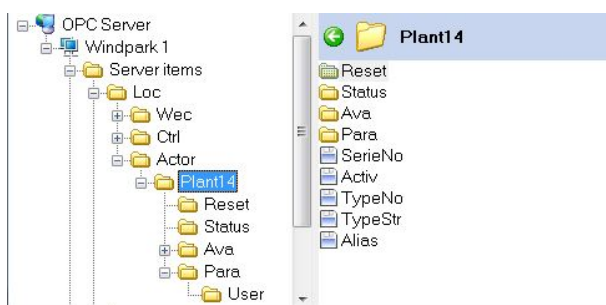


Fig. 27: Loc/Actor/Plant<N> node for PDI

Node/Item	Description
Reset	Control system reset (WEC)
Status	PDI status data
Log	PDI data logger
Ava	PDI availability data
Para	PDI parameters
SerieNo	Serial number assigned by ENERCON
Active	WEC is in active or passive mode
TypeNo	ENERCON control system type number
TypeStr	ENERCON control system type string
Alias	Identification selectable by customer

3.7.1.1 The Loc/Actor/Plant<N>/Para node for PDI

The following parameters are available for the ENERCON SCADA PDI:

Item	Description	Unit
1000	Rated Power	kW
1001	Rated Reactive power	Kvar
1010	Partial P Limitation	kW
4000	Flex P Step 1	
4001	Flex P Step 2	
4002	Flex P Step 3	
4003	Flex P Step 4	
4004	Flex P Step 5	
4005	Flex P Step 6	
4006	Flex P Step 7	
4007	Flex P Step 8	
4010	Flex Phi Step 1	
4011	Flex Phi Step 2	
4012	Flex Phi Step 3	
4013	Flex Phi Step 4	
4014	Flex Phi Step 5	
4015	Flex Phi Step 6	
4016	Flex Phi Step 7	
4017	Flex Phi Step 8	
4050	Signal Fault Behav.	
4120	P Adjustment Offset	%
4121	P Adjustment Scale	
4130	Phi Adjustment Offs.	deg.
4131	Phi Adjustment Scale	
4140	Cos Phi Adj. Offset	
4141	Cos Phi Adj. Scale	
4150	Q Adjustment Offset	%
4151	Q Adjustment Scale	
4200	AO 5 Clamp 1 Type	
4201	AO 5 Clamp 1 Min.	
4202	AO 5 Clamp 1 Max.	
4203	AO 5 Clamp 1 Scale	

Item	Description	Unit
4210	AO 5 Clamp 2 Type	
4211	AO 5 Clamp 2 Min.	
4212	AO 5 Clamp 2 Max.	
4213	AO 5 Clamp 2 Scale	
4300	Flex P Type	
4301	Flex Phi Type	
4350	P Default Set Point	%
4351	Phi Default Set P.	deg.
4352	Q Default Set Point	%
4360	P Display Set Point	%
4361	Phi Display Set P.	deg.
4362	Q Display Set Point	%
5002	Phi Minimum	deg.
5003	Phi Maximum	deg.

4 Nodes and items for closed-loop and open-loop control

In OPC, signs (+ and –) are used in accordance with IEC 61400-21.

4.1 Session items for access for control purposes

Any access to the ENERCON PDI-OPC for control purposes is performed on the basis of a standardised mechanism. Therefore, with all control actions, the session items below exist in the data tree of the respective control value.

Item	Description
Session State	The current status of a session (Short Integer/Read Access) 0: Session free 1: Session reserved 2: Session parameter input 3: Waiting period in loop mode 4: Wait session end 5: Session blocked 108: Busy 109: Access denied 121: Value error 174: Wrong user ID 175: Insufficient access rights
Session Timeout	Shows the remaining time of a session timeout (Integer/Read Access)
Public Key	Key generated by the server for access to that particular session (Long Word/Read Access)
SessionRequest	Array for requesting a session reservation (Array 3 Long Word/Read Access). The array contains the following values in the sequence defined below: Session ID Random value by client which client can use to determine successful reservation. This value can be read by the clients. User ID The user ID represents a user in the SCADA System. SCADA users and their authorisations are created in the SCADA System by ENERCON (Service IT). This value cannot be read and is therefore invisible to other clients. Private Key The Private Key is a random value and a subkey for access to the respective session. This value cannot be read and is therefore invisible to other clients.
SessionSubmit	Array for running a session (2 Long Word/Read & Write Access). The array contains the following values in the sequence defined below: Private Key The private key registered with the server when logging on to the session. This value cannot be read and is therefore invisible to other clients. Public Key The public key made available by the server after logging on to the session.

4.1.1 Session items for access to individual WECs for control purposes

The *Loc/Wec/Plant<N>/Ctrl* node contains items for access to individual WECs for control purposes

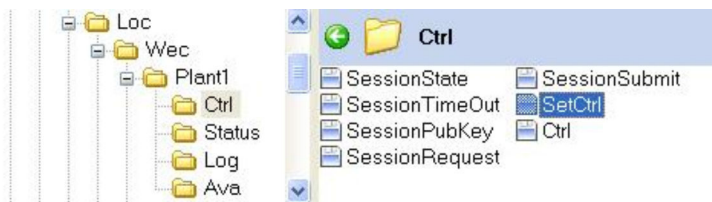


Fig. 28: Session items for access to individual WECs for control purposes

Data is transmitted in a single session.

Item	Description
Ctrl	The current status of the wind energy converter (Short Integer/Read Access) 0: Turbine in operation 1: WEC stopped at 60° 2: WEC stopped at 90° 129: WEC stopped at 60° (with higher priority). Cannot be restarted. 130: WEC stopped at 90° (with higher priority). Cannot be restarted. 255: Value has not yet been initialised. Fault in communication with WEC Only operating states with values 0–2 can be changed.
SetCtrl	Array for setting a control value (Array of 3 Long Word / Read & Write Access) The array contains the following values in the sequence defined below: Control value 1 = WEC Stop (60°) 2 = WEC Stop (90°) 0 = WEC Start Private Key The private key selected by the user at the start of the session. Public Key The public key made available by the server at the start of the session.

Timings

Normal timeout during a session	60 s
Extended timeout after setting a parameter/control value	60 s
Session end timeout (delay of new reservation)	180 s
Error timeout (after three incorrect entries)	180 s
User ID error timeout	300 s

4.1.2 Session items for wind farm start/stop of all WECs in a wind farm

Wind farm control of WECs is performed taking into account the current control statuses of individual WECs. A consistent status cannot be determined across all WECs and can therefore not be displayed. For safe operation, the control status of individual WECs should be evaluated before performing any switching operation.

The wind farm control items reside in the `/Loc/Wec/Ctrl` node.

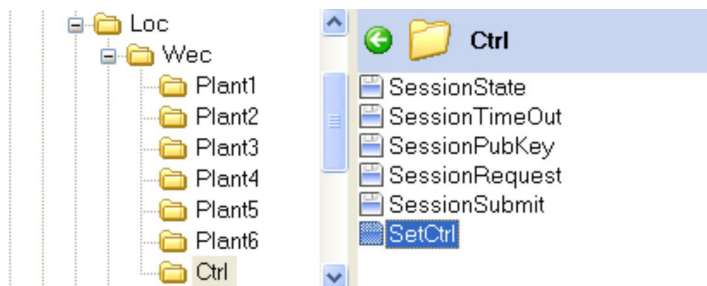


Fig. 29: Session items for wind farm start/stop of all WECs in a wind farm

Data is transmitted in a single session.

Item	Description
SetCtrl	Array for setting a control value (Array of 3 Long Word / Read & Write Access) The array contains the following values in the sequence defined below: Control value 1 = Stop wind farm (60°) 2 = Stop wind farm (90°) 0 = Start wind farm/Put wind farm into operation The control value can also be retrieved for review. Private Key The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key The public key made available by the server after logging on to the session.

Timings

Normal timeout during a session	60 s
Extended timeout after setting a parameter/control value	60 s
Session end timeout (delay of new reservation)	180 s
Error timeout (after three incorrect entries)	180 s
User ID error timeout	300 s

4.1.3 Session items for resetting the control system of individual wind energy converters

The *Loc/Wec/Plant<N>/Reset* node contains the items for resetting the control system of individual wind energy converters.

Data is transmitted in a single session.

Item	Description
SetReset	Array for setting a control value. (Array of 3 Long Word / Read & Write Access) The array contains the following values in the sequence defined below: WEC number Private Key The private key selected by the user at the start of the session Public Key The public key made available by the server at the start of the session

Timings

Normal timeout during a session	60 s
Extended timeout after setting a parameter/ control value	60 s
Session end timeout (delay of new reservation)	180 s
Error timeout (after three incorrect entries)	180 s
User ID error timeout	300 s

4.2 Items for wind farm closed-loop and open-loop control (ENERCON SCADA Server)



The items provided through SCADA PDI-OPC are based on the SCADA control system; i.e., the item path and the items contained in it change, depending on the SCADA control settings. Interpretation of the items depends on the respective control system type.

4.2.1 Weekly table control

Weekly table control (not a control system) specifying the wind farm's relative power set-point (0-100 %) and/or the phase angle of the power injected is performed in a single session. The relevant items reside the */Scada/TableCtr1* node.

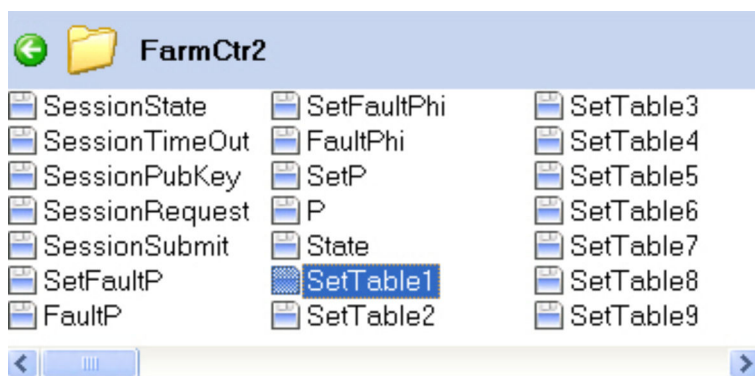


Fig. 30: Weekly table control items

Item	Description
TableX	Current table entry X of 40 for specifying power and phase angle setpoints (array of 8 Double/Read Access). The array contains the following values in the sequence defined below: Weekday 1 = Sunday through 7 = Saturday Hour Hour = 0 to 23 Minute Minute = 0 to 59 Second Second = 0 to 59 Power setpoint Relative power setpoint [%] = 0 to 100 Phase angle setpoint Absolute phase angle setpoint [%]
SetTableX	Array for setting table entry X of 40 for specifying power and phase angle setpoints (array of 6 Double/Read Access). The array contains the following values in the sequence defined below: Weekday 1 = Sunday through 7 = Saturday Hour Hour = 0 to 23 Minute Minute = 0 to 59 Second Second = 0 to 59 Power setpoint Relative power setpoint [%] = 0 to 100 Phase angle setpoint Absolute phase angle setpoint [°] Private Key The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key The public key made available by the server after logging on to the session.
	Table entries are deleted by writing Day of week = / Hour = / Minute = / Second = 0.

Timings

Normal timeout during a session	60 s
Extended timeout after setting a parameter/control value	60 s
Session end timeout (delay of new reservation)	300 s
Error timeout (after three incorrect entries)	300 s
User ID error timeout	300 s

4.2.2 Farm Control 2

Closed-loop wind farm control data with the measuring point at the point of common coupling and setpoint specification in the SCADA system are transmitted in a single session. Optionally, weekly table control can be activated for setpoint specification.

The absolute power setpoint of the control system will then result from the relative table value and the absolute power setpoint.

$$P_{\text{control}} = (P_{\text{abs}} \times P_{\text{table}}) \div 100$$

Data are transmitted in a single session. The relevant items reside in the *Scada/Farmctr2* node.

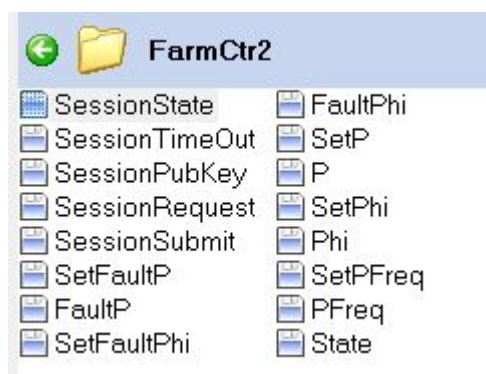


Fig. 31: Farm Control 2 items

Item	Description
State	Status of closed-loop control system. 0 = Normal operation
FaultP	Current relative power default value during a fault (e.g. communication fault) of the controller (Double/Read Access)
SetFaultP	Array for setting the power default value (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Default power value Relative default power value [%]. The set value can also be retrieved for review. Private Key The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key The public key made available by the server after logging on to the session.

Item	Description
FaultPhi	Current absolute phase angle default value during a fault (e.g. communication fault) of the controller (Array of 3 Double/Read & Write Access)
SetFaultPhi	Array for setting the phase angle default value (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Phase angle default value Absolute phase angle default value [°]. The set value can also be retrieved for review. Private Key The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key The public key made available by the server after logging on to the session.
P	Absolute power setpoint (Double/Read Access)
SetP	Array for setting the power setpoint (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Power setpoint Absolute phase angle power setpoint [kW]. The set value can also be retrieved for review. Private Key The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key The public key made available by the server after logging on to the session.
Phi	Current absolute phase angle setpoint [°] (Double/Read) This item will not exist if weekly table control is active.
SetPhi	Array for setting the phase angle setpoint [°] (Array of 3 Double/Read & Write Access). This item will not exist if weekly table control is active. The array contains the following values in the sequence defined below: Phase angle setpoint Phase angle setpoint [°] Private Key The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key The public key made available by the server after logging on to the session.

Item	Description
SetPFreq	Array for setting the frequency reserve capacity (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Power setpoint SetPFreq[0] Relative frequency reserve capacity control value [%] The set value can also be read for control purposes. Private key SetPFreq[1] The private key with which the client server was registered when logging on to the session. This value cannot be read and is therefore invisible to other clients. Public key SetPFreq[2] The public key made available by the server after logging on to the session.
PFreq	Current relative frequency reserve capacity setpoint (Double/Read Access)

Timings

Normal timeout during a session	60 s
Extended timeout after setting a parameter/control value	60 s
Session end timeout (delay of new reservation)	300 s
Error timeout (after three incorrect entries)	300 s
User ID error timeout	300 s

4.2.3 Apparent power control 2

Closed-loop wind farm control data with the measuring point at the point of common coupling and setpoint specification in the SCADA system is transmitted in a single session. Optionally, weekly table control can be activated for setpoint specification.

The absolute power setpoint of the control system will then result from the relative table value and the absolute power setpoint.

$$P_{\text{control}} = (P_{\text{abs}} \times P_{\text{table}}) \div 100$$

The apparent power setpoint is an apparent power limit. Reactive power (Phi) will be reduced once this limit has been reached.

All data points of Farm Control 2 are also present with Apparent Power Control 2. Only the apparent power limit is added.

The relevant items reside in the */Scada/Appctr2* node.

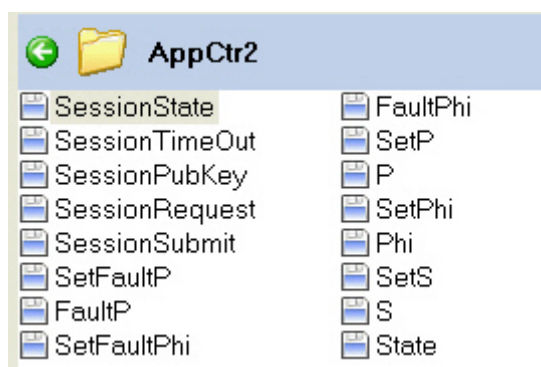


Fig. 32: Apparent power control 2 items

Item	Description
S	Absolute apparent power setpoint limit (Double/Read Access)
SetS	Array for setting the apparent power limit (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Apparent power limit Absolute apparent power limit [kVA]. The set value can also be retrieved for review. Private Key The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key The public key made available by the server after logging on to the session.

Timings

Normal timeout during a session	60 s
Extended timeout after setting a parameter/ control value	60 s
Session end timeout (delay of new reservation)	300 s
Error timeout (after three incorrect entries)	300 s
User ID error timeout	300 s

4.2.4 Open Loop Control (OIctr1)

Open Loop Control is not a control system but a wind farm control. The setpoints specified via OPC are checked, limited and passed on to the WEC in a steady cycle. No additional hardware (RTU or FCU) is required for open-loop control by means of OPC. With this type of control, the OPC (and thus also the client connection) is integrated into the regulator cycle and monitored.

Data is transmitted in a loop session. The relevant items reside in the */Scada/OIctr1* node.

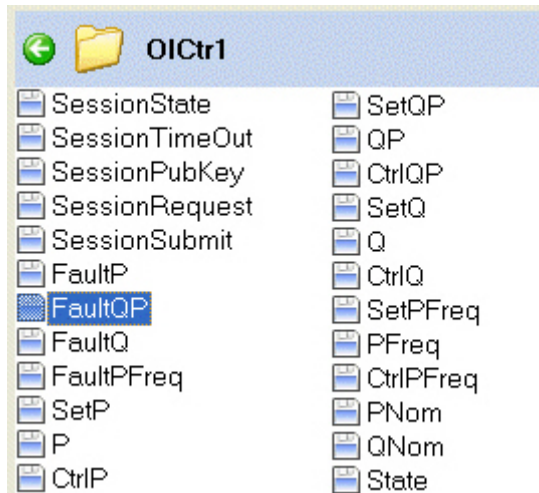


Fig. 33: Open Loop Control (OIctr1) items

Item	Description
FaultP	Current relative default value during a fault of the regulator (Double/Read Access)
FaultQP	Current absolute Q/P default value during a fault of the control system (Double/Read Access)
FaultQ	Current relative Q default value during a fault of the control system (Double/Read Access)
FaultPFreq	Current relative frequency reserve capacity default value during a fault of the control system (Double/Read Access)
SetP	Array for setting the power setpoint (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Power setpoint SetP [0] Relative power setpoint [%] The set value can also be retrieved for review. Private Key SetP [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetP [2] The public key made available by the server after logging on to the session.
P	Current relative power setpoint (Double/Read Access)

Item	Description
CtrlP	Current relative control value P (Double/Read Access)
SetQP	Array for setting the Q/P setpoint (Array of 3 Double/Read & Write Access). This array contains the following values in the sequence defined below: Power setpoint SetQP [0] Q/P setpoint The set value can also be retrieved for review. Private Key SetQP [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetQP [2] The public key made available by the server after logging on to the session.
QP	Current absolute Q/P setpoint (Double/Read Access)
CtrlQP	Current control value Q/P (Double/Read Access)
SetQ	Array for setting the reactive power setpoint (Array of 3 Double/Read & Write Access). This array contains the following values in the sequence defined below: Power setpoint SetQ [0] Relative reactive power setpoint [%]. The set value can also be retrieved for review. Private Key SetQ [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetQ [2] The public key made available by the server after logging on to the session.
Q	Current relative reactive power setpoint (Double/Read Access)
CtrlQ	Current relative control value Q (Double/Read Access)

Item	Description
SetPFreq	Array for setting the frequency reserve capacity (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Power setpoint SetPFreq [0] Relative frequency reserve capacity control value [%]. The set value can also be retrieved for review. Private Key SetPFreq [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetPFreq [2] The public key made available by the server after logging on to the session.
PFreq	Current relative frequency reserve capacity setpoint (Double/Read Access)
CtrlPFreq	Current frequency reserve capacity control value (Double/Read Access)
PNom	Rated power of the wind farm (Double/Read Access)
QNom	Rated reactive power of the wind farm (Double/Read Access)
State	Status of closed-loop control system. 0 = Normal operation

Timings

Normal timeout during a session	10 × regular cycle
Extended timeout after setting a parameter/control value	10 × regular cycle
Session end timeout (delay of new reservation)	60 s
Error timeout (after 3 incorrect entries)	300 s
User ID error timeout	300 s

4.2.5 Manual control 1 (ManCtr1)

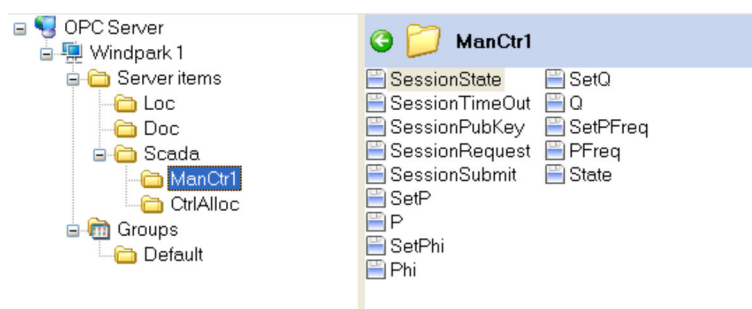


Fig. 34: Items for Manual Control 1

Item	Description
SetP	Array for setting the power setpoint (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Power setpoint SetP [0] Relative power setpoint [%] The set value can also be retrieved for review. Private Key SetP [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetP [2] The public key made available by the server after logging on to the session.
P	Current relative power setpoint (Double/Read Access)
SetPhi	Array for configuring the phase angle setpoint (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Phase angle setpoint SetPhi [0] Phase angle setpoint in ° The set value can also be retrieved for review. Private Key SetPhi [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetPhi [2] The public key made available by the server after logging on to the session.
Phi	Current phase angle setpoint (Double/Read Access)

Item	Description
SetQ	Array for setting the reactive power setpoint (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Reactive power setpoint SetQ [0] Relative reactive power setpoint [%]. The set value can also be retrieved for review. Private Key SetQ [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetQ [2] The public key made available by the server after logging on to the session.
Q	Current relative reactive power setpoint (Double/Read Access)
SetPFreq	Array for setting the frequency reserve capacity setpoint (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Power setpoint SetPFreq [0] Relative frequency reserve capacity setpoint [%] The set value can also be retrieved for review. Private Key SetPFreq [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetPFreq [2] The public key made available by the server after logging on to the session.
PFreq	Current relative power setpoint (Double/Read Access)
State	Open-loop control system status. 0 = Normal operation

Timings

Normal timeout during a session	60 s
Extended timeout after setting a parameter/control value	60 s
Session end timeout (delay of new reservation)	300 s
Error timeout (after 3 incorrect entries)	300 s
User ID error timeout	300 s

4.2.6 Power Consumption Management

The Power Consumption Management data is transmitted in a single session. The relevant items reside in the */SCADA/Pcm<N>* node.

A node with the relevant items is created for each active group of WECs. The number of *SetPlant* and *Plant* items correspond to the number of WECs in the group.

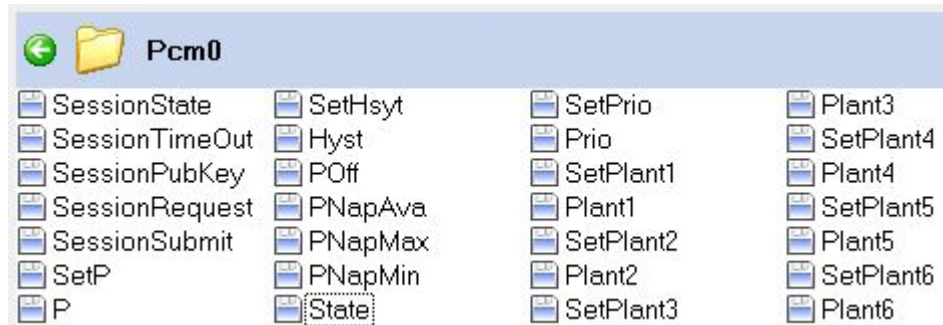


Fig. 35: Items for Power Consumption Management (six WECs in the group)

Item	Description
SetP	Array for setting the power consumption setpoint (Array of 3 Long Word / Read & Write Access). The array contains the following values in the sequence defined below: Power consumption setpoint SetP [0] Power consumption setpoint in kW The set value can also be retrieved for review. Private Key SetP [1] The private key which the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetP [2] The public key made available by the server after logging on to the session.
P	Current power consumption setpoint (Integer/Read Access)

Item	Description
SetHyst	Array for setting the hysteresis (Array of 3 Long Word/Read & Write Access). The array contains the following values in the sequence defined below: Hysteresis setpoint SetHyst [0] Hysteresis setpoint in % The set value can also be retrieved for review. Private Key SetHyst [1] The private key which the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetHyst [2] The public key made available by the server after logging on to the session.
Hyst	Current hysteresis (Integer/Read Access)
POff	Current available power (Integer/Read Access)
PNapAva	Average power consumption measured at the network connection point (NCP) or determined by a virtual network connection point (Integer/Read Access)
PNapMax	Maximum power consumption measured at the network connection point or determined by a virtual network connection point (integer/Read Access)
PNapMin	Maximum power consumption measured at the network connection point or determined by a virtual network connection point (Integer/Read Access)
State	Status of Power Consumption Management 0 = Program start consumption 1 = Program end consumption 60 = Parameters changed 61 = WEC priority changed 62 = WEC mode changed 105 = Error sensor data logger 215 = Error WEC measurement value 246 = Error NCP sensor measurement value

Item	Description
SetPrio	Array for setting the priority (array variable value of n+2 Long Word/Read & Write Access; n = number of WECs in the group). The array contains the following values in the sequence defined below: Places in the list (1 to n) SetPrio [0] to SetPrio [n-1] WEC number of the WECs The set values can also be retrieved for review. Private Key SetPrio [n] The private key which the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetPrio [n+1] The public key made available by the server after logging on to the session.
Prio	Current set priority or call sequence/list (array with variable value of n Word/Read Access; number of WECs in the group). Prio[0] = WEC 1, Prio [1] = WEC 2, etc.

Item	Description
SetPlant<N>	Array for selecting the WEC operating mode (Array of 4 Long Word/Read & Write Access). The array contains the following values in the sequence defined below: Operating mode SetPlant<N> [0] Operating mode of WEC Power Consumption Management. 0 = Automatic mode 1 – 253 = Manual mode; value corresponds to duration in hours The set value can also be retrieved for review. Power consumption availability SetPlant<N> [1] Power consumption availability for manual WEC operating mode. The set value can also be retrieved for review. Private Key SetPlant<N> [2] The private key which the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetPlant<N> [3] The public key made available by the server after logging on to the session.
Plant<N>	Array for reading current WEC information (Array of 8 Long Word/Read Access). The array contains the following values in the sequence defined below: Maintenance Plant<N> [0] 1 = WEC in maintenance Error code Plant<N> [1] Operating mode Plant<N> [2] Deactivated = WEC is logged off Power Consumption Management (e.g. for maintenance purposes) Sync = Synchronisation, e.g. after restarting the WEC Error = Communication problem (read only) Current operating mode of Power Consumption Management in the respective WEC. Time Plant<N> [3] Remaining duration of manual mode. Power consumption availability Plant<N> [4] Current power consumption availability. Reserved Plant<N> [5] Current WEC power reserve. Reserved (t-1) Plant<N> [6] Reserved power consumption of previous cycle.

Item	Description
	Desired power Plant<N> [7] Power required by WEC.

Timings

Regular timeout during a session	60 s
Extended timeout after setting a parameter/ control value	60 s
Session end timeout (delay of new reservation)	300 s
Error timeout (after 3 incorrect entries)	300 s
User ID error timeout	300 s

4.3 Items for RTU/FCU closed-loop and open-loop control

The items described below are exchanged between OPC XML DA Server and the connected peripheral SCADA units (RTU, FCU). Individual control functions are (de)activated directly on the peripheral SCADA units.

Please refer to the device documentation for further information about the scope of functionality and the supported data points.

The *Loc/Ctrl/Plant<N>/Ctrl* node contains items for RTU or FCU control access.

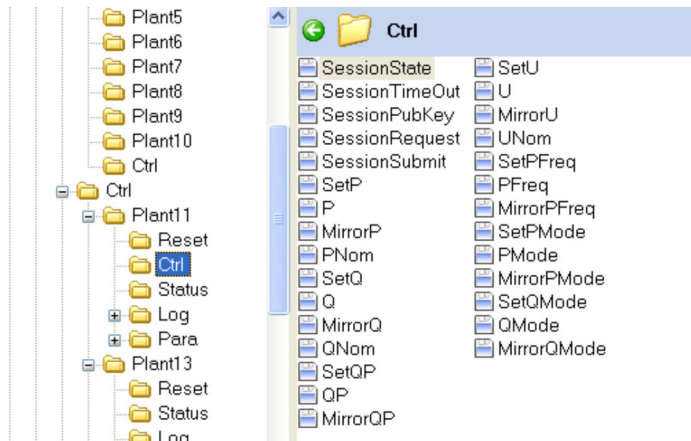


Fig. 36: *Loc/Ctrl/Plant<N>/Ctrl* node for FCU or RTU



Alternatively, the power factor ($\cos \phi$) or the phase angle ϕ or Q/P can be transmitted as the reactive power control setpoint. This setting has to be made on the OPC server.

Item	Description
SetP	Array for setting the active power setpoint (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Power setpoint SetP [0] Relative active power setpoint [%]. Minimum value: 0 % Maximum value: 100 % 100 % means no reduction. The set value can also be retrieved for review. Private Key SetP [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetP [2] The public key made available by the server after logging on to the session.
P	Relative active power setpoint – Return (RTU/FCU) [% of P_{nom} for wind farm] (Double/Read Access)
MirrorP	Active power setpoint – OPC return (Double/Read Access)
PNom	Rated power of the wind farm [kW] (Double/Read Access)
SetQ	Array for setting the reactive power setpoint (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Reactive power setpoint SetQ [0] Relative reactive power setpoint [%]. Minimum value: -100 % Maximum value: +100 % 0 % means no reactive power. The set value can also be retrieved for review. Private Key SetQ [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetQ [2] The public key made available by the server after logging on to the session.
Q	Reactive power setpoint – Return (RTU / FCU) [% of P_{nom} for wind farm] (Double/Read Access)
MirrorQ	Reactive power setpoint – OPC return (Double/Read Access)
QNom	Rated reactive power of the wind farm [kvar] (Double/Read Access)

Item	Description
SetCos(Phi)* Alternatively: SetQ/P, SetPhi	Array for setting the power factor setpoint (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Power factor setpoint SetCos(Phi) [0] Minimum value: -1 Maximum value: +1 0 means no reactive power. The set value can also be retrieved for review. Private Key SetCos(Phi) [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetCos(Phi) [2] The public key made available by the server after logging on to the session. Phase angle setpoint Phi Minimum value: -90 ° Maximum value: +90 ° 0 ° means no reactive power. The setup and name of the respective array corresponds to the rest of the setpoints. Power factor setpoint Q/P Minimum value: -3 Maximum value: +3 0 means no reactive power. The setup and name of the respective array corresponds to the rest of the setpoints
Cos(Phi) Alternatively: Q/P, Phi	Power factor setpoint – Return (RTU/FCU) [% of P_{nom} for wind farm] (Double/Read Access)
MirrorCos(Phi) Alternatively: MirrorQP, MirrorPhi	Power factor setpoint – OPC return (Double/Read Access)

Item	Description
SetU	Array for setting the voltage setpoint (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Voltage setpoint SetU [0] Relative voltage setpoint [%]. Minimum value: 0 % Maximum value: 200 % 100 % means rated voltage. The set value can also be retrieved for review. Private Key SetU [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetU [2] The public key made available by the server after logging on to the session.
U	Voltage setpoint – Return (RTU / FCU) [% of P_{nom} for wind farm] (Double/Read Access)
MirrorU	Voltage setpoint – OPC return (Double/Read Access)
UNom	Nominal voltage of the wind farm [V] (Double/Read Access)
SetPFreq	Array for setting the frequency reserve capacity (Array of 3 Double/Read & Write Access). The array contains the following values in the sequence defined below: Power setpoint SetPFreq [0] Relative frequency reserve capacity control value [%]. Minimum value: 50 % Maximum value: 100 % 100 % means no reserve capacity. The set value can also be retrieved for review. Private Key SetPFreq [1] The private key the client uses to log on to the server for the session. This value cannot be read and is therefore invisible to other clients. Public Key SetPFreq [2] The public key made available by the server after logging on to the session.
PFreq	Frequency reserve capacity setpoint – Return (RTU / FCU) [% of P_{nom} for wind farm] (Double/Read Access)
MirrorPFreq	frequency reserve capacity setpoint – OPC return (Double/Read Access)
SetPMode	Setpoint for active power control mode
PMode	Setpoint for active power control mode – Mirror (RTU/FCU)

Item	Description
MirrorPMode	Setpoint for active power control mode – Mirror (OPC)
SetQMode	Setpoint for reactive power control mode
QMode	Setpoint for reactive power control mode – Mirror (RTU/FCU)
MirrorQMode	Setpoint for reactive power control mode – Mirror (OPC)

Timings

Normal timeout during a session	60 s
Extended timeout after setting a parameter/ control value	60 s
Session end timeout (delay of new reservation)	60 s
Error timeout (after 3 incorrect entries)	300 s
User ID error timeout	300 s

5 Overview of items and abbreviations used

The use of the names of the items provided by the OPC server is the same as with the ENERCON SCADA System.

5.1 Wind energy converters

The following items reside in the */Loc/Wec/Plant<N>/Log/Wecstd* node:

Item	Description
arwAbW	Generated energy [kWh]
arwAbWorkH	Production time [h]
arwAbWrkM	Production time [min]
lrwSmpNRot	Minimum rotor speed [1/min]
lrwSmpP	Minimum power [kW]
lrwSmpQ	Minimum reactive power [kvar]
lrwSmpVWi	Minimum wind speed [m/s]
mrwAbGoPos	Average nacelle position [°]
mrwSmpNRot	Average rotor speed [1/min]
mrwSmpP	Average power [kW]
mrwSmpPext*	Average available active power, limited by external setpoints [kW]
mrwSmpPfm*	Average of maximum available active power, limited by force majeure [kW]
mrwSmpPte*	Average of technically available active power [kW]
mrwSmpPwin*	Average of theoretically available active power in the wind [kW]
mrwSmpQ	Average reactive power [kvar]
mrwSmpVWi	Average wind speed [m/s]
prwSmpNRot	Maximum rotor speed [1/min]
prwSmpP	Maximum power [kW]
prwSmpQ	Maximum reactive power [kvar]
prwSmpVWi	Maximum wind speed [m/s]

* For a detailed description, see ch. 3.2, p. 9.

The following items reside in the */Loc/Wec/Plant<N>/Log/T40* node:

Item	Description
mr40A0	Stator temperature
mr40A1	Rotor temperature
mr40A2	Nacelle ambient temperature
mr40A3	Front bearing temperature
mr40A4	Rear bearing temperature
mr40A8	Tower temperature
mr40A9	Control cabinet temperature
mr40A10	Rectifier cabinet 1 temperature [°C]
mr40A11	Rectifier cabinet 2 temperature [°C]
mr40A19	Blade A temperature
mr40A20	Blade B temperature
mr40A21	Blade C temperature

The following items reside in the */Loc/Wec/Plant<N>/Log/T44* node:

Item	Description
mr44A0	Stator temperature
mr44A1	Rotor temperature
mr44A2	Nacelle ambient temperature
mr44A3	Nacelle temperature
mr44A4	Front bearing temperature
mr44A5	Rear bearing temperature
mr44A6	Blade A temperature
mr44A7	Blade B temperature
mr44A8	Blade C temperature
mr44A12	Nacelle control cabinet temperature
mr44A13	Control cabinet temperature
mr44A14	Tower temperature
mr44A18	Transformer temperature
mr44A24	Ambient temperature
mr44A25	Control cabinet yaw converter temperature

The following items reside in the */Loc/Wec/Plant<N>/Log/T44b* node:

Item	Description
mr44bA0	Stator temperature
mr44bA1	Rotor temperature
mr44bA2	Nacelle ambient temperature
mr44bA3	Nacelle temperature
mr44bA4	Front bearing temperature
mr44bA5	Rear bearing temperature
mr44bA6	Blade A temperature
mr44bA7	Blade B temperature
mr44bA8	Blade C temperature
mr44bA12	Nacelle control cabinet temperature
mr44bA13	Control cabinet temperature
mr44bA14	Tower temperature
mr44bA18	Transformer temperature
mr44bA28	Ambient temperature
mr44bA29	Control cabinet yaw converter temperature

The following items reside in the */Loc/Wec/Plant<N>/Log/T48* node:

Item	Description
mr48A0	Spinner temperature
mr48A1	Front bearing temperature
mr48A2	Rear bearing temperature
mr48A6	Blade A pitch heat sink
mr48A7	Blade B pitch heat sink
mr48A8	Blade C pitch heat sink
mr48A9	Blade A pitch control box
mr48A10	Blade B pitch control box
mr48A11	Blade C pitch control box
mr48A12	Blade A temperature
mr48A13	Blade B temperature
mr48A14	Blade C temperature
mr48A15	Rotor temperature 1
mr48A16	Rotor temperature 2
mr48A17	Stator temperature 1

Item	Description
mr48A18	Stator temperature 2
mr48A19	Nacelle ambient temperature
mr48A20	Nacelle temperature
mr48A21	Nacelle control cabinet temperature
mr48A22	Rectifier 1 heat sink
mr48A23	Rectifier 2 heat sink
mr48A24	Rectifier cabinet temperature
mr48A25	Excitation heat sink
mr48A34	Ambient temperature
mr48A35	Tower temperature
mr48A36	Control cabinet temperature
mr48A37	Transformer temperature
mr48A46	Sys 1. temperature Power cabinet 1
mr48A55	Sys 1. temperature Power cabinet 2
mr48A64	Sys 1. temperature Power cabinet 3

The following items reside in the */Loc/Wec/Plant<N>/Log/T66* node:

Item	Description
mr66A0	Stator temperature
mr66A1	Rotor temperature
mr66A2	Nacelle ambient temperature
mr66A3	Nacelle temperature
mr66A4	Spinner temperature
mr66A5	Front bearing temperature
mr66A6	Rear bearing temperature
mr66A7	Blade A temperature
mr66A8	Blade B temperature
mr66A9	Blade C temperature
mr66A19	Nacelle control cabinet temperature
mr66A20	Control cabinet temperature
mr66A21	Tower temperature
mr66A25	Transformer temperature
mr66A26	Temperature – Sys. Power cabinet 1
mr66A27	Temperature – Sys. Power cabinet 2

Item	Description
mr66A28	Temperature – Sys. Power cabinet 3
mr66A29	Temperature – Sys. Power cabinet 4
mr66A30	Temperature – Sys. Power cabinet 5
mr66A31	Temperature – Sys. Power cabinet 6
mr66A45	Ambient temperature

The following items reside in the */Loc/Wec/Plant<N>/Log/T70* node:

Item	Description
mr70A0	Stator temperature
mr70A1	Rotor temperature
mr70A2	Nacelle ambient temperature
mr70A3	Nacelle temperature
mr70A4	Spinner temperature
mr70A5	Front bearing temperature
mr70A6	Rear bearing temperature
mr70A7	Blade A temperature
mr70A8	Blade B temperature
mr70A9	Blade C temperature
mr70A19	Nacelle control cabinet temperature
mr70A20	Control cabinet temperature
mr70A21	Tower temperature
mr70A25	Transformer temperature
mr70A39	Ambient temperature
mr70A40	Control cabinet yaw converter temperature

The following items reside in the */Loc/Wec/Plant<N>/Log/T70b* node:

Item	Description
mr70bA0	Stator temperature
mr70bA1	Rotor temperature
mr70bA2	Nacelle ambient temperature
mr70bA3	Nacelle temperature
mr70bA4	Spinner temperature
mr70bA5	Front bearing temperature
mr70bA6	Rear bearing temperature

Item	Description
mr70bA7	Blade A temperature
mr70bA8	Blade B temperature
mr70bA9	Blade C temperature
mr70bA19	Nacelle control cabinet temperature
mr70bA20	Control cabinet temperature
mr70bA21	Tower temperature
mr70bA25	Transformer temperature
mr70bA51	Ambient temperature
mr70bA61	Control cabinet yaw converter temperature

The following items reside in the */Loc/Wec/Plant<N>/Log/T82a1* node:

Item	Description
mr82A0	Spinner temperature
mr82A1	Front bearing temperature
mr82A2	Rear bearing temperature
mr82A6	Blade A pitch heat sink temperature
mr82A7	Blade B pitch heat sink temperature
mr82A8	Blade C pitch heat sink temperature
mr82A9	Blade A pitch control cabinet temperature
mr82A10	Blade B pitch control cabinet temperature
mr82A11	Blade C pitch control cabinet temperature
mr82A12	Blade A temperature
mr82A13	Blade B temperature
mr82A14	Blade C temperature
mr82A15	Rotor temperature 1
mr82A16	Rotor temperature 2
mr82A17	Stator temperature 1
mr82A18	Stator temperature 2
mr82A19	Nacelle ambient temperature
mr82A20	Nacelle temperature
mr82A21	Nacelle control cabinet temperature
mr82A22	Rectifier 1 heat sink temperature
mr82A23	Rectifier 2 heat sink temperature
mr82A24	Rectifier cabinet temperature

Item	Description
mr82A25	Excitation heat sink temperature
mr82A36	Ambient temperature
mr82A37	Tower temperature
mr82A38	Control cabinet temperature
mr82A39	Transformer temperature

The following items reside in the */Loc/Wec/Plant<N>/Log/T82a2* node:

Item	Description
mr82A48	Sys 1. temperature Power cabinet 1
mr82A57	Sys 1. temperature Power cabinet 2
mr82A66	Sys 1. temperature Power cabinet 3
mr82A75	Sys 1. temperature Power cabinet 4
mr82A84	Sys 1. temperature Power cabinet 5
mr82A93	Sys 1. temperature Power cabinet 6
mr82A102	Sys 1. temperature Power cabinet 7
mr82A111	Sys 1. temperature Power cabinet 8
mr82A120	Sys 1. temperature Power cabinet 9
mr82A129	Sys 1. temperature Power cabinet 10
mr82A138	Sys 1. temperature Power cabinet 11

The following items reside in the */Loc/Wec/Plant<N>/Log/T101a1* node:

Item	Description
mr101A0	Spinner temperature
mr101A1	Front bearing temperature
mr101A2	Rear bearing temperature
mr101A13	Blade A pitch control cabinet temperature
mr101A14	Blade B pitch control cabinet temperature
mr101A15	Blade C pitch control cabinet temperature
mr101A16	Blade A temperature
mr101A17	Blade B temperature
mr101A18	Blade C temperature
mr101A19	Rotor temperature 1
mr101A20	Rotor temperature 2
mr101A21	Stator temperature 1

Item	Description
mr101A22	Stator temperature 2
mr101A27	Nacelle ambient temperature 1
mr101A28	Nacelle ambient temperature 2
mr101A29	Nacelle temperature
mr101A30	Nacelle control cabinet temperature
mr101A31	Main carrier temperature
mr101A36	Control cabinet rectifier temperature
mr101A38	Control cabinet yaw converter temperature
mr101A54	Control cabinet fan converter temperature
mr101A63	Ambient temperature
mr101A64	Tower temperature
mr101A65	Control cabinet temperature
mr101A66	Transformer temperature

The following items reside in the */Loc/Wec/Plant<N>/Log/T101a2* node:

Item	Description
mr101A75	Temperature – Sys. 1; Power cabinet 1
mr101A84	Temperature – Sys. 1; Power cabinet 2
mr101A93	Temperature – Sys. 1; Power cabinet 3
mr101A102	Temperature – Sys. 1; Power cabinet 4
mr101A111	Temperature – Sys. 1; Power cabinet 5
mr101A120	Temperature – Sys. 1; Power cabinet 6
mr101A129	Temperature – Sys. 1; Power cabinet 7
mr101A138	Temperature – Sys. 2; Power cabinet 1
mr101A147	Temperature – Sys. 2; Power cabinet 2
mr101A156	Temperature – Sys. 2; Power cabinet 3
mr101A165	Temperature – Sys. 2; Power cabinet 4
mr101A174	Temperature – Sys. 2; Power cabinet 5
mr101A183	Temperature – Sys. 2; Power cabinet 6
mr101A192	Temperature – Sys. 2; Power cabinet 7

The following items reside in the */Loc/Wec/Plant<N>/Log/T126a1* node:

Item	Description
mr112A0	Spinner temperature
mr112A2	Front bearing temperature
mr112A3	Rear bearing temperature
mr112A17	Blade A pitch control cabinet temperature
mr112A18	Blade B pitch control cabinet temperature
mr112A19	Blade C pitch control cabinet temperature
mr112A20	Blade A temperature 1
mr112A21	Blade A temperature 2
mr112A22	Blade B temperature 1
mr112A23	Blade B temperature 2
mr112A24	Blade C temperature 1
mr112A25	Blade C temperature 2
mr112A26	Rotor temperature 1
mr112A27	Rotor temperature 2
mr112A28	Stator temperature 1.1
mr112A29	Stator temperature 2.1
mr112A40	Control cabinet rectifier temperature 1.1
mr112A41	Control cabinet rectifier temperature 1.2
mr112A42	Control cabinet rectifier temperature 2.1
mr112A43	Control cabinet rectifier temperature 2.2
mr112A44	Control cabinet rectifier temperature 3.1
mr112A45	Control cabinet rectifier temperature 3.2
mr112A46	Control cabinet rectifier temperature 4.1
mr112A47	Control cabinet rectifier temperature 4.2
mr112A48	Nacelle ambient temperature 1
mr112A49	Nacelle temperature
mr112A50	Nacelle control cabinet temperature
mr112A55	Control cabinet excitation temperature
mr112A56	Control cabinet yaw converter temperature
mr112A301	Control cabinet fan converter temperature
mr112A302	Nacelle ambient temperature 2
mr112A64	Ambient temperature
mr112A65	Tower temperature

Item	Description
mr112A66	Control cabinet temperature
mr112A67	Transformer system 1 temperature
mr112A68	Transformer system 2 temperature
mr112A69	Transformer system 3 temperature
mr112A70	Transformer system 4 temperature

The following items reside in the */Loc/Wec/Plant<N>/Log/T126a2* node:

Item	Description
mr112A79	Temperature – Sys. 1; Power cabinet 1
mr112A88	Temperature – Sys. 1; Power cabinet 2
mr112A97	Temperature – Sys. 1; Power cabinet 3
mr112A106	Temperature – Sys. 1; Power cabinet 4
mr112A115	Temperature – Sys. 1; Power cabinet 5
mr112A259	Temperature – Sys. 1; Power cabinet 6
mr112A315	Temperature – Sys. 1; Power cabinet 7
mr112A324	Temperature – Sys. 1; Power cabinet 8
mr112A124	Temperature – Sys. 2; Power cabinet 1
mr112A133	Temperature – Sys. 2; Power cabinet 2
mr112A142	Temperature – Sys. 2; Power cabinet 3
mr112A151	Temperature – Sys. 2; Power cabinet 4
mr112A160	Temperature – Sys. 2; Power cabinet 5
mr112A268	Temperature – Sys. 2; Power cabinet 6
mr112A333	Temperature – Sys. 2; Power cabinet 7
mr112A342	Temperature – Sys. 2; Power cabinet 8

The following items reside in the */Loc/Wec/Plant<N>/Log/T126a3* node:

Item	Description
mr112A169	Temperature – Sys. 3; Power cabinet 1
mr112A178	Temperature – Sys. 3; Power cabinet 2
mr112A187	Temperature – Sys. 3; Power cabinet 3
mr112A196	Temperature – Sys. 3; Power cabinet 4
mr112A205	Temperature – Sys. 3; Power cabinet 5
mr112A277	Temperature – Sys. 3; Power cabinet 6
mr112A351	Temperature – Sys. 3; Power cabinet 7

Item	Description
mr112A360	Temperature – Sys. 3; Power cabinet 8
mr112A214	Temperature – Sys. 4; Power cabinet 1
mr112A223	Temperature – Sys. 4; Power cabinet 2
mr112A232	Temperature – Sys. 4; Power cabinet 3
mr112A241	Temperature – Sys. 4; Power cabinet 4
mr112A250	Temperature – Sys. 4; Power cabinet 5
mr112A286	Temperature – Sys. 4; Power cabinet 6
mr112A269	Temperature – Sys. 4; Power cabinet 7
mr112A378	Temperature – Sys. 4; Power cabinet 8

5.2 ENERCON GDA

The following items reside in the */Loc/Ctrl/Plant<N>/Log/Farmctr1* node:

Item	Description (GDA)
arucAbQc2	Reactive power measurement, capacitive [kvarh] (Active power consumption)
arucAbQcp	Reactive power measurement, capacitive [kvarh] (Active power production)
arucAbQi2	Reactive power measurement, inductive [kvarh] (Active power consumption)
arucAbQin	Reactive power measurement, inductive [kvarh] (Active power production)
arucAbWcm	Energy consumption [kWh]
arucAbWpr	Energy production [kWh]
lrucSmpCos	Minimum power factor $\cos\phi$
lrucSmpFre	Minimum frequency [Hz]
lrucSmpl1	Minimum current L1 [A]
lrucSmpl2	Minimum current L2 [A]
lrucSmpl3	Minimum current L3 [A]
lrucSmpP	Minimum power [kW]
lrucSmpQ	Minimum reactive power [kvar]
lrucSmpV1	Minimum voltage U12 [V]
lrucSmpV2	Minimum voltage U23 [V]
lrucSmpV3	Minimum voltage U31 [V]
mrucSmpCos	Average power factor $\cos\phi$
mrucSmpFre	Average frequency [Hz]
mrucSmpl1	Average current L1 [A]
mrucSmpl2	Average current L2 [A]
mrucSmpl3	Average current L3 [A]
mrucSmpP	Average power [kW]
mrucSmpQ	Average reactive power [kvar]
mrucSmpV1	Average voltage U12 [V]
mrucSmpV2	Average voltage U23 [V]
mrucSmpV3	Average voltage U31 [V]
prucSmpCos	Maximum power factor $\cos\phi$
prucSmpFre	Maximum frequency [Hz]
prucSmpl1	Maximum current L1 [A]

Item	Description (GDA)
prucSmpI2	Maximum current L2 [A]
prucSmpI3	Maximum current L3 [A]
prucSmpP	Maximum power [kW]
prucSmpQ	Maximum reactive power [kvar]
prucSmpV1	Maximum voltage U12 [V]
prucSmpV2	Maximum voltage U23 [V]
prucSmpV3	Maximum voltage U31 [V]

5.3 ENERCON METEO

The following items reside in the */Loc/Meteo/Plant<N>/Log/Wind1* node:

Item	Description (wind measurement)
lrmSmpV1	Minimum wind speed 1 [m/s]
lrmSmpV2	Minimum wind speed 2 [m/s]
lrmSmpV3	Minimum wind speed 3 [m/s]
lrmSmpV4	Minimum wind speed 4 [m/s]
lrmSmpV51	Minimum horizontal wind speed [m/s] (Ultrasonic anemometer)
lrmSmpV52	Minimum vertical wind speed [m/s] (Ultrasonic anemometer)
mrmSmpPo1	WIND: -----
mrmSmpPo2	WIND: -----
mrmSmpPx1	Average X wind direction 1 [°]
mrmSmpPx2	Average X wind direction 2 [°]
mrmSmpPx5	Average X wind direction [°] (Ultrasonic anemometer)
mrmSmpPy1	Average Y wind direction 1 [°]
mrmSmpPy2	Average Y wind direction 2 [°]
mrmSmpPy5	Average Y wind direction [°] (Ultrasonic anemometer)
mrmSmpV1	Average wind speed 1 [m/s]
mrmSmpV2	Average wind speed 2 [m/s]
mrmSmpV3	Average wind speed 3 [m/s]
mrmSmpV4	Average wind speed 4 [m/s]
mrmSmpV51	Average horizontal wind speed [m/s] (Ultrasonic anemometer)
mrmSmpV52	Average vertical wind speed [m/s] (Ultrasonic anemometer)
prmSmpV1	Maximum wind speed 1 [m/s]
prmSmpV2	Maximum wind speed 2 [m/s]
prmSmpV3	Maximum wind speed 3 [m/s]
prmSmpV4	Maximum wind speed 4 [m/s]
prmSmpV51	Maximum horizontal wind speed [m/s] (Ultrasonic anemometer)
prmSmpV52	Maximum vertical wind speed [m/s] (Ultrasonic anemometer)

The following items reside in the /Loc/Meteo/Plant<N>/Log/Weather2 node:

Item	Description (weather measurement)
mrmtSmpTm2	Average temperature [°C]
mrmtSmpPr2	Average air pressure [hPa]
mrmtSmpRa2	Average precipitation [mm/min]
mrmtSmpSu2	Average solar radiation [W/m ²]
mrmtSmpWe2	Average air humidity [%]

5.4 ENERCON Type 1 SCU

The following items reside in the */Loc/Ctrl/Plant<N>/Log/Substat1* node:

Item	Description (Type 1 SCU)
mruuSmpHV1	Average high voltage 1 U12 [V]
mruuSmpHV2	Average high voltage 1 U23 [V]
mruuSmpHV3	Average high voltage 1 U31 [V]
mruuSmpMV1	Average medium voltage 2 U12 [V]
mruuSmpMV2	Average medium voltage 2 U23 [V]
mruuSmpMV3	Average medium voltage 2 U31 [V]
mruuSmpP	Average power [kW]
mruuSmpQ	Average reactive power [kvar]

5.5 ENERCON Type 2 SCU

The following items reside in the */Loc/Ctrl/Plant<N>/Log/Subphase2* node:

Item	Description (Type 2 SCU)
aruwAbQcp	Reactive power measurement, capacitive [kvarh] (Active power production)
aruwAbQin	Reactive power measurement, inductive [kvarh] (Active power production)
aruwAbWcm	Energy consumption [kWh]
aruwAbWkH	Production time [h]
aruwAbWpr	Energy production [kWh]
lruwSmpCos	Minimum power factor $\cos\phi$
lruwSmpFre	Minimum frequency [Hz]
lruwSmpHI1	Minimum current 1 L1 [A] (high-voltage side)
lruwSmpHI2	Minimum current 1 L2 [A] (high-voltage side)
lruwSmpHI3	Minimum current 1 L3 [A] (high-voltage side)
lruwSmpHV1	Minimum high voltage 1 U12 [V]
lruwSmpHV2	Minimum high voltage 1 U23 [V]
lruwSmpHV3	Minimum high voltage 1 U31 [V]
lruwSmpMI1	Minimum current 2 L1 [A] (medium-voltage side)
lruwSmpMI2	Minimum current 2 L2 [A] (medium-voltage side)
lruwSmpMI3	Minimum current 2 L3 [A] (medium-voltage side)
lruwSmpMV1	Minimum medium voltage 2 U12 [V]
lruwSmpMV2	Minimum medium voltage 2 U23 [V]

Item	Description (Type 2 SCU)
lruwSmpMV3	Minimum medium voltage 2 U31 [V]
lruwSmpP	Minimum power [kW]
lruwSmpQ	Minimum reactive power [kvar]
lruwSmpS	Minimum apparent power [kVA]
mruwSmpCos	Average power factor $\cos\phi$
mruwSmpFre	Average frequency [Hz]
mruwSmpHI1	Average current 1 L1 [A] (high-voltage side)
mruwSmpHI2	Average current 1 L2 [A] (high-voltage side)
mruwSmpHI3	Average current 1 L3 [A] (high-voltage side)
mruwSmpHV1	Average high voltage 1 U12 [V]
mruwSmpHV2	Average high voltage 1 U23 [V]
mruwSmpHV3	Average high voltage 1 U31 [V]
mruwSmpMI1	Average current 2 L1 [A] (medium-voltage side)
mruwSmpMI2	Average current 2 L2 [A] (medium-voltage side)
mruwSmpMI3	Average current 2 L3 [A] (medium-voltage side)
mruwSmpMV1	Average medium voltage 2 U12 [V]
mruwSmpMV2	Average medium voltage 2 U23 [V]
mruwSmpMV3	Average medium voltage 2 U31 [V]
mruwSmpP	Average power [kW]
mruwSmpQ	Average reactive power [kvar]
mruwSmpS	Average apparent power [kVA]
pruwSmpCos	Maximum power factor $\cos\phi$
pruwSmpFre	Maximum frequency [Hz]
pruwSmpHI1	Maximum current 1 L1 [A] (high-voltage side)
pruwSmpHI2	Maximum current 1 L2 [A] (high-voltage side)
pruwSmpHI3	Maximum current 1 L3 [A] (high-voltage side)
pruwSmpHV1	Maximum high voltage 1 U12 [V]
pruwSmpHV2	Maximum high voltage 1 U23 [V]
pruwSmpHV3	Maximum high voltage 1 U31 [V]
pruwSmpMI1	Maximum current 2 L1 [A] (medium-voltage side)
pruwSmpMI2	Maximum current 2 L2 [A] (medium-voltage side)
pruwSmpMI3	Maximum current 2 L3 [A] (medium-voltage side)
pruwSmpMV1	Maximum medium voltage 2 U12 [V]
pruwSmpMV2	Maximum medium voltage 2 U23 [V]

Item	Description (Type 2 SCU)
pruwSmpMV3	Maximum medium voltage 2 U31 [V]
pruwSmpP	Maximum power [kW]
pruwSmpQ	Maximum reactive power [kvar]
pruwSmpS	Maximum apparent power [kVA]

5.6 ENERCON SCU with temperature measurement

Only for projects using the ENERCON SCU with temperature measuring equipment. Allocation of items to measuring points in the transmission substation is project-specific.

The following items reside in the */Loc/Ctrl/Plant<n>/Log/SubTemp2* node:

Item	Description (SCU with temperature measurement)
lruwSmpT1	Minimum transformer temperature 1 [°C]
lruwSmpT2	Minimum transformer temperature 2 [°C]
lruwSmpT3	Minimum transformer temperature 3 [°C]
lruwSmpT4	Minimum transformer temperature 4 [°C]
lruwSmpT5	Minimum room temperature 1 [°C]
lruwSmpT6	Minimum room temperature 2 [°C]
lruwSmpT7	Minimum room temperature 3 [°C]
lruwSmpT8	Minimum room temperature 4 [°C]
lruwSmpT9	Minimum outside temperature 1 [°C]
lruwSmpT10	Minimum outside temperature 2 [°C]
lruwSmpT11	Minimum free temperature 1 [°C]
lruwSmpT12	Minimum free temperature 2 [°C]
mruwSmpT1	Average transformer temperature 1 [°C]
mruwSmpT2	Average transformer temperature 2 [°C]
mruwSmpT3	Average transformer temperature 3 [°C]
mruwSmpT4	Average transformer temperature 4 [°C]
mruwSmpT5	Average room temperature 1 [°C]
mruwSmpT6	Average room temperature 2 [°C]
mruwSmpT7	Average room temperature 3 [°C]
mruwSmpT8	Average room temperature 4 [°C]
mruwSmpT9	Average outside temperature 1 [°C]
mruwSmpT10	Average outside temperature 2 [°C]
mruwSmpT11	Average free temperature 1 [°C]
mruwSmpT12	Average free temperature 2 [°C]
pruwSmpT1	Maximum transformer temperature 1 [°C]
pruwSmpT2	Maximum transformer temperature 2 [°C]
pruwSmpT3	Maximum transformer temperature 3 [°C]
pruwSmpT4	Maximum transformer temperature 4 [°C]
pruwSmpT5	Maximum room temperature 1 [°C]
pruwSmpT6	Maximum value Room temperature 2 [°C]

Item	Description (SCU with temperature measurement)
pruwSmpT7	Maximum room temperature 3 [°C]
pruwSmpT8	Maximum room temperature 4 [°C]
pruwSmpT9	Maximum outside temperature 1 [°C]
pruwSmpT10	Maximum outside temperature 2 [°C]
pruwSmpT11	Maximum free temperature 1 [°C]
pruwSmpT12	Maximum free temperature 2 [°C]

5.7 ENERCON FCU

The following items reside in the */Loc/Ctrl/Plant<N>/Log/Fcuctr1* node:

Item	Description (FCU)
lrfcSmpCos	Minimum power factor $\cos\phi$
lrfcSmpFre	Minimum frequency [Hz]
lrfcSmpl1	Minimum current L1 [A]
lrfcSmpl2	Minimum current L2 [A]
lrfcSmpl3	Minimum current L3 [A]
lrfcSmpP	Minimum power [kW]
lrfcSmpQ	Minimum reactive power [kvar]
lrfcSmpV1	Minimum voltage U1N [V]
lrfcSmpV2	Minimum voltage U2N [V]
lrfcSmpV3	Minimum voltage U3N [V]
mrfcSmpCos	Average power factor $\cos\phi$
mrfcSmpFre	Average frequency [Hz]
mrfcSmpl1	Average current L1 [A]
mrfcSmpl2	Average current L2 [A]
mrfcSmpl3	Average current L3 [A]
mrfcSmpP	Average power [kW]
mrfcSmpQ	Average reactive power [kvar]
mrfcSmpS1	Average active power setpoint 1 [kW]
mrfcSmpS2	Average active power setpoint 2 [kW]
mrfcSmpS3	Average active power setpoint 3 [kW]
mrfcSmpV1	Average voltage U1N [V]
mrfcSmpV2	Average voltage U2N [V]
mrfcSmpV3	Average voltage U3N [V]
prfcSmpCos	Maximum power factor $\cos\phi$
prfcSmpFre	Maximum frequency [Hz]
prfcSmpl1	Maximum current L1 [A]
prfcSmpl2	Maximum current L2 [A]
prfcSmpl3	Maximum current L3 [A]
prfcSmpP	Maximum power [kW]
prfcSmpQ	Maximum reactive power [kvar]
prfcSmpV1	Maximum voltage U1N [V]

Item	Description (FCU)
prfcSmpV2	Maximum voltage U2N [V]
prfcSmpV3	Maximum voltage U3N [V]

5.8 ENERCON RTU

The following items reside in the */Loc/Ctrl/Plant<N>/Log/Rtu1* node:

Item	Description (Rtu1)
aruiAbQcap	Reactive power measurement, capacitive [kvar] (active power generation)
aruiAbQin	Reactive power measurement, inductive [kvar] (active power generation)
aruiAbWcm	Energy consumption [kWh]
aruiAbWkH	Production time [h]
aruiAbWpr	Energy production [kWh]
lruiSmpCos	Minimum power factor $\cos\phi$
lruiSmpFre	Minimum frequency [Hz]
lruiSmpI1	Minimum current 1 L1 [A]
lruiSmpI2	Minimum current 1 L2 [A]
lruiSmpI3	Minimum current 1 L3 [A]
lruiSmpP	Minimum power [kW]
lruiSmpQ	Minimum reactive power [kvar]
lruiSmpS	Minimum apparent power [kVA]
lruiSmpU1	Minimum voltage 1 U12 [V]
lruiSmpU2	Minimum voltage 1 U23 [V]
lruiSmpU3	Minimum voltage 1 U31 [V]
mruiSmpCos	Average power factor $\cos\phi$
mruiSmpFre	Average frequency [Hz]
mruiSmpI1	Average current 1 L1 [A]
mruiSmpI2	Average current 1 L2 [A]
mruiSmpI3	Average current 1 L3 [A]
mruiSmpP	Average power [kW]
mruiSmpQ	Average reactive power [kvar]
mruiSmpS	Average apparent power [kVA]
mruiSmpS1	Average active power setpoint 1 [kW]
mruiSmpS2	Average active power setpoint 2 [kW]
mruiSmpS3	Average active power setpoint 3 [kW]
mruiSmpU1	Average voltage 1 U12 [V]
mruiSmpU2	Average voltage 1 U23 [V]
mruiSmpU3	Average voltage 1 U31 [V]
pruiSmpCos	Maximum power factor $\cos\phi$

Item	Description (Rtu1)
pruiSmpFre	Maximum frequency [Hz]
pruiSmpl1	Maximum current 1 L1 [A]
pruiSmpl2	Maximum current 1 L2 [A]
pruiSmpl3	Maximum current 1 L3 [A]
pruiSmpP	Maximum power [kW]
pruiSmpQ	Maximum reactive power [kvar]
pruiSmpS	Maximum apparent power [kVA]
pruiSmpU1	Maximum voltage 1 U12 [V]
pruiSmpU2	Maximum voltage 1 U23 [V]
pruiSmpU3	Maximum voltage 1 U31 [V]

The following items reside in the /Loc/Ctrl/Plant<N>/Log/Rtuphas1 node:

Item	Description (Rtuphas1)
lruiSmpP1	Minimum power L1 [kW]
lruiSmpP2	Minimum power L2 [kW]
lruiSmpP3	Minimum power L3 [kW]
lruiSmpQ1	Minimum reactive power L1 [kvar]
lruiSmpQ2	Minimum reactive power L2 [kvar]
lruiSmpQ3	Minimum reactive power L3 [kvar]
mruiSmpP1	Average power L1 [kW]
mruiSmpP2	Average power L2 [kW]
mruiSmpP3	Average power L3 [kW]
mruiSmpQ1	Average reactive power L1 [kvar]
mruiSmpQ2	Average reactive power L2 [kvar]
mruiSmpQ3	Average reactive power L3 [kvar]
pruiSmpP1	Maximum power L1 [kW]
pruiSmpP2	Maximum power L2 [kW]
pruiSmpP3	Maximum power L3 [kW]
pruiSmpQ1	Maximum reactive power L1 [kvar]
pruiSmpQ2	Maximum reactive power L2 [kvar]
pruiSmpQ3	Maximum reactive power L3 [kvar]

5.9 Solar energy converter

The following items reside in the *Loc/Wec/Conv/Plant<N>/Log/Solar1* node:

Item	Description
arsoW	Energy produced [kWh]
arsoWorkH	Operating hours [h]
lrsoFreq	Minimum frequency [Hz]
lrsoI	Minimum current [A]
lrsoIntens	Minimum solar irradiance
lrsoL1I	Minimum current L1 [A]
lrsoL1U	Minimum voltage, Level U12 [V]
lrsoL2I	Minimum current L2 [A]
lrsoL2U	Minimum voltage, Level U23 [V]
lrsoL3I	Minimum current L3 [A]
lrsoL3U	Minimum voltage, Level U31 [V]
lrsoP	Minimum power [kW]
lrsoU1	Minimum voltage, Level 1 [V]
lrsoU2	Minimum voltage, Level 2 [V]
mrsoFreq	Average frequency [Hz]
mrsoI	Average current [A]
mrsoIntens	Average solar irradiance
mrsoL1I	Average current L1 [A]
mrsoL1U	Average voltage, Level U12 [V]
mrsoL2I	Average current L2 [A]
mrsoL2U	Average voltage, Level U23 [V]
mrsoL3I	Average current L3 [A]
mrsoL3U	Average voltage, Level U31 [V]
mrsoP	Average power [kW]
mrsoU1	Average voltage, Level 1 [V]
mrsoU2	Average voltage, Level 2 [V]
prsoFreq	Maximum frequency [Hz]
prsoI	Maximum current [A]
prsoIntens	Maximum solar irradiance
prsoL1I	Maximum current L1 [A]
prsoL1U	Maximum voltage, Level U12 [V]
prsoL2I	Maximum current L2 [A]

Item	Description
prsoL2U	Maximum voltage, Level U23 [V]
prsoL3I	Maximum current L3 [A]
prsoL3U	Maximum voltage, Level U31 [V]
prsoP	Maximum power [kW]
prsoU1	Maximum voltage, Level 1 [V]
prsoU2	Maximum voltage, Level 2 [V]

5.10 Medium-voltage data acquisition

The following items reside in the *Loc/Sens/Plant<N>/Log/Mspgerf1* node:

Item	Description
lrsmSmpFre	Minimum frequency [Hz]
lrsmSmpU1	Minimum voltage U12 [V]
lrsmSmpU2	Minimum voltage U23 [V]
lrsmSmpU3	Minimum voltage U31 [V]
mrsmSmpFre	Average frequency [Hz]
mrsmSmpU1	Average voltage U12 [V]
mrsmSmpU2	Average voltage U23 [V]
mrsmSmpU3	Average voltage U31 [V]
prsmSmpFre	Maximum frequency [Hz]
prsmSmpU1	Maximum voltage U12 [V]
prsmSmpU2	Maximum voltage U23 [V]
prsmSmpU3	Maximum voltage U31 [V]

Glossary of terms

ENERCON SCADA System	The ENERCON SCADA system performs data acquisition, remote monitoring, and open-loop/closed-loop control both for individual wind energy converters and entire wind farms. It enables customers and ENERCON Service to monitor operating states and to analyse stored operating data.
FCU	The ENERCON Farm Control Unit (FCU) serves to control specific electrical parameters (e.g. active power, power factor, voltage) at the wind farm's PoC.
GDA	The Grid Data Acquisition system (GDA) is used for measuring key electrical variables at the wind farm's point of common coupling. These values are transmitted to the ENERCON SCADA Server for display and control purposes as well as data storage.
Item	Within the context of OPC, the combination of a (physical) value and its properties.
METEO	The ENERCON METEO system serves to acquire and analyse meteorological data in conjunction with ENERCON SCADA.
OPC XML DA	Open Process Connectivity (see www.opcfoundation.org). Extensible Markup Language (XML) is used, among other things, for the exchange of data between computer systems, in particular over the Internet. DA means "Data Access".
PDI	The Process Data Interface (PDI) is an interface between different communication devices, for example between the ENERCON SCADA system and the grid operator. ENERCON offers different PDIs for connecting the ENERCON SCADA system to external communication points.
RTU-C	ENERCON SCADA Remote Terminal Unit-Control (RTU-C) is a superordinate closed-loop wind farm control system. The RTU-C uses the output values registered at the point of connection to achieve the specified setpoints.
SCU	The Substation Control Unit (SCU) is a communication interface between the control system of the transmission substation and the ENERCON SCADA system.
Server	A server is a program that communicates with other programs and can provide special services and data to the latter.